# Carbonatite-hosted REE prospectivity under extreme label scarcity in Canada: national-scale sampling and transferability, and a semi-supervised deviation network in three geological provinces

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**Abstract**

Carbonatites are the primary geological sources of rare earth elements (REEs), yet their prospectivity mapping is constrained by extreme label scarcity, the diverse geological settings hosting these deposits, the absence of reliable negative labels and limited model interpretability. Recent national-scale models report test AUCs above 0.9, but use augmented labels, negatives from the ground least similar to the occurrences and random splits. This study tests that design, then whether a semi-supervised anomaly-detection model, the Deviation Network (DevNet), offers advantages over Random Forest (RF), XGBoost and positive–unlabelled learners in the Superior Province, with the Cordilleran Orogen and Grenville Province as contrasts, and whether the provincial models transfer between provinces. Geophysical rasters and their derivatives, proximity-to-structure features and H3-grid geology were extracted per province. All models were trained on the occurrences together with an identical 5,000-point background sample, evaluated on a disjoint random-background set with pooled out-of-fold average precision (AP) and area under the ROC curve (AUC) under repeated random and 50–200 km spatially blocked cross-validation, and interpreted with SHapley Additive exPlanations (SHAP). Rebuilt on national data, the earlier design yields a test AUC of 1.000 with both of its classifiers, which falls to 0.903 (AP 0.292) for its random forest once labels are not augmented and negatives are drawn at random, and models do not transfer between provinces. In Superior, DevNet achieves the highest AP under both random (0.38) and spatially blocked (0.37–0.22) cross-validation, and the regional ordering (Superior > Cordillera > Grenville) agrees with previous studies. Carbonatite prospectivity models should therefore be built and validated within individual provinces, where semi-supervised methods match the reality that unexplored ground is not proven barren.

**Keywords:** Mineral prospectivity mapping, Rare earth elements, Carbonatite, Deviation network, Random Forest, Positive–unlabelled learning, Spatial cross-validation

## 1 Introduction

Rare earth elements (REEs) are critical to low-carbon energy, electronics and defence technologies, yet economic concentrations are geologically uncommon (Wang et al., 2020). Most REE resources are hosted by carbonatites and associated alkaline intrusions, whose distinctive but partly overlapping geophysical and geochemical signatures make them natural targets for data-driven mineral prospectivity mapping

(MPM; Simandl & Paradis, 2018; Verplanck et al., 2014). In Canada, however, confirmed carbonatite-hosted REE occurrences are few and spatially clustered (Fig. 2): after near-duplicates are merged, a single geological province holds between about ten and thirty of them.

Two properties of this problem undermine standard supervised classification. First, positives are extremely scarce: mineralisation is a rare event (Xiong & Zuo, 2018), known occurrences are only a partial and biased sample of the mineral system (Coolbaugh et al., 2007; Parsa & Carranza, 2021; Zuo et al., 2021), and under $k$-fold cross-validation each fold holds only a handful of them, so any estimate is dominated by which occurrences happen to be held out. Second, there are no true negatives. Ground without a recorded occurrence is unexplored rather than barren, and designating it negative is an arbitrary modelling choice whose outcome depends on which background points are drawn (Zuo & Wang, 2020). Together these properties motivate positive–unlabelled (PU) and semi-supervised anomaly-detection formulations, in which mineralised sites are treated as rare anomalies relative to a large, mostly barren but unlabelled population (Bekker & Davis, 2020; Y. Chen & Wu, 2019; Luo et al., 2026; Xiong & Zuo, 2016, 2021).

The Deviation Network (DevNet; Pang et al., 2019) is a semi-supervised deep anomaly detector that learns an anomaly score from a few labelled anomalies and a large unlabelled pool, pulling the scores of unlabelled data towards a Gaussian reference while pushing labelled anomalies a fixed margin above it. Luo et al. (2026) adapted it to MPM (DEEP-SEAM), encoding geology as distances to mapped units and adding neighbourhood statistics of the evidence layers. Whether such a model outperforms discriminative baselines when positives number in the low tens has not been tested under evaluation designs that control for the specific ways in which small, clustered label sets inflate scores. The closest prior study, a national-scale model of Canadian carbonatite-hosted REE ± Nb deposits (Parsa et al., 2024), reported test AUCs of 0.91–0.96 with random forests and convolutional neural networks (CNNs), the highest for a CNN. Its labels were augmented by replacing each occupied S2 grid cell with its four child cells, its negatives were drawn from the ground least similar to the positives, and it was evaluated by stratified random 70/30 splits of the augmented labels. A follow-up study recommended positive augmentation with labels generated by generative adversarial networks (GANs) for the same commodity (Parsa et al., 2025). Each of these design choices can make the held-out test easier than prospecting new ground. A national model also answers a different question from a provincial one. Pooling every occurrence of a country raises the label count, but it mixes occurrences from geological provinces whose geophysical settings, and whose barren ground, differ completely; whether a model trained on such a pool predicts within any one province, and whether one province's occurrences say anything about another's, is rarely tested (Meyer & Pebesma, 2021; Ploton et al., 2020).

This study starts at the national scale. We rebuild the S2-grid design of Parsa et al. (2024) on national data and compare it with a design that keeps one label per occurrence, draws background at random and evaluates at a realistic prevalence. Because carbonatites formed in different geological settings, whose expression in the evidence layers differs, the main part of the study models provinces separately: the Superior Province of the Canadian Shield, the province in which labels are scarcest relative to area, with the Cordilleran Orogen and Grenville Province modelled with the identical pipeline as contrasting cases. All analyses share one occurrence table and one input rule; inputs whose value depends on the extent of the study area are computed per geological province, so that a point has the same value nationally and provincially. We ask: (i) how much of the high AUC of the national S2-grid design comes from its sampling and augmentation? (ii) within Superior, does DevNet outperform tree-based learners (RF and XGBoost) and positive–unlabelled learners once every model sees the same data and folds, and does any advantage survive spatially blocked validation? (iii) how much of the skill comes from geological map classes that mark known complexes? (iv) can positive augmentation compensate for label scarcity? and (v) does a model trained in one province predict another? We close by interpreting the provincial models with SHapley Additive exPlanations (SHAP), using a rank-based rule for the direction of each feature's effect, and by checking the Superior maps against an untrained REE resource and the mapped Kapuskasing Structural Zone.

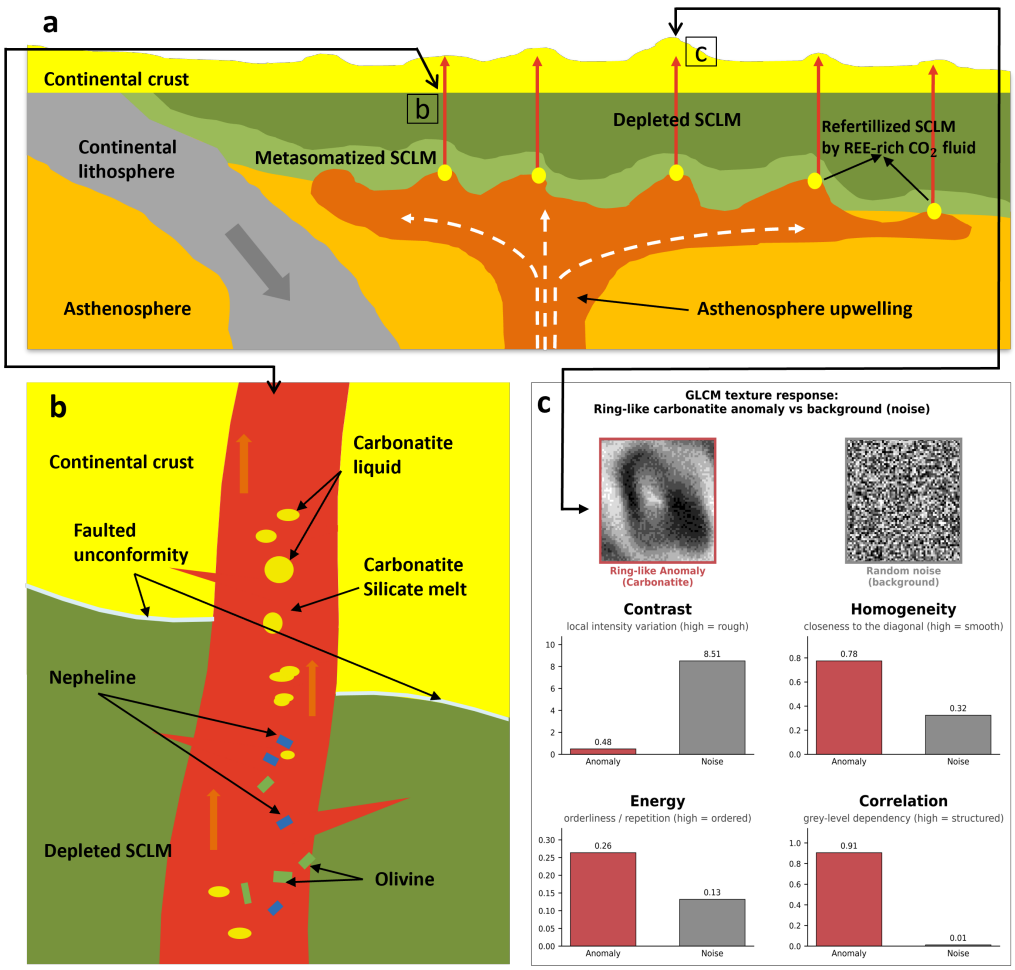



**Fig. 1** Source–transport–trap model for carbonatite-hosted REE systems and its geophysical proxies. (a) Source: refertilised subcontinental lithospheric mantle near the LAB at the craton margin, indexed by LAB depth and margin proximity; panel modified after Hou et al. (2023). (b) Transport along trans-lithospheric faults, indexed by fault and worm proximity and Moho depth; modified after Yaxley et al. (2021). (c) Trap: GLCM texture response of a ring-like anomaly and of random noise.

# 2 Geological setting and data

## 2.1 Carbonatite-hosted REE systems and their geophysical expression

Carbonatites occur predominantly in Precambrian cratonic areas, and individual provinces record repeated emplacement over hundreds of millions of years, implying long-lived, episodically reactivated trans-lithospheric pathways (Woolley & Bailey, 2012). A component of their REE endowment is attributed to subcontinental lithospheric mantle refertilised by REE- and $CO_2$-rich metasomatic fluids (Hou et al., 2023; Yaxley et al., 2021). Each element of this source–transport–trap system has a mappable geophysical proxy (Fig. 1). Depth to the lithosphere–asthenosphere boundary (LAB) indexes the thickness and edge geometry of the cratonic root that hosts the fertile mantle source, which exerts first-order control on magmatic ore systems (Griffin et al., 2013; Groves & Santosh, 2021; Hoggard et al., 2020); Moho depth indexes crustal thickness; and proximity to faults and to the loci of maximum horizontal gravity and magnetic gradient ("worms"; Hornby et al., 1999) proxies the structures that channel ascending melts. At the emplacement site, dense and magnetic alkaline–carbonatite complexes produce positive gravity anomalies and intense circular to annular magnetic anomalies (Drenth, 2014; Thomas et al., 2016). Grey-level co-occurrence matrix (GLCM) textures summarise the local two-dimensional heterogeneity produced by such concentric patterns without assuming a target diameter (Cooper & Cowan, 2005b; Hall-Beyer, 2017), unlike explicit radial-symmetry detectors tuned to ring-complex size (Aranha et al., 2021).

## 2.2 Study areas

The national study area is the union of the onshore geological provinces, orogens and platforms of the Geological Map of Canada (Map 1860A; Wheeler et al., 1996), without the continental shelves and oceanic crust (12.3 million km$^2$). The three provincial study areas are the Superior Province, the Grenville Province and the Cordilleran Orogen polygons of the same map. Carbonatite-hosted occurrences of the Phanerozoic, accreted Cordillera differ from those of the Canadian Shield in their geophysical and geological characteristics (Parsa et al., 2024), which suggests that a single cross-terrane model would not transfer; Section 4.1 tests this directly. Windows of one province enclosed by another and smaller than 2,000 km$^2$, such as the Churchill-basement windows of the Monashee and Malton complexes in the Cordillera, were assigned to the surrounding province.

The Superior Province is the world's largest Archean craton. Carbonatite–alkaline magmatism in the Canadian Shield clusters into episodes at about 2.68 Ga, 1.90–1.88 Ga and 1.16–1.08 Ga (Rukhlov & Bell, 2010), and in the Superior Province many of the complexes are aligned along crustal-scale structures, most prominently the Kapuskasing Structural Zone (KSZ; Gittins et al., 1967; Percival & West, 1994). The 15 Superior occurrences used here (Fig. 2b) comprise a cluster around the Montviel and Lac Shortt carbonatites in the eastern Abitibi region of Québec (Deposit/Prospect and Showing class) and occurrences dispersed across the southern and western Superior Province, including Nemegosenda Lake, Lackner Lake, Kapuskasing, Martison Lake, Prairie Lake and Carb Lake. The Superior Province thus combines the fewest occurrences per unit area of the three provinces with a spatial structure, one tight cluster plus scattered singletons, in which a model can either learn the cluster or learn what the scattered occurrences have in common.

## 2.3 Occurrence labels

Positive labels were compiled from the carbonatite-hosted records of the USGS global REE occurrence database (Orris et al., 2018) and the occurrence table of Parsa et al. (2024), totally 102 records across Canada. Alkaline-hosted occurrences were excluded after a preliminary leave-block-out classification showed them to be geophysically separable from carbonatite-hosted ones. Records within 2 km of each other were merged by single linkage, so that drill holes or zones of one deposit, listed once or in both sources, count once; without this step near-identical points from one deposit fall into both training and validation folds. Each occurrence keeps the development class of Parsa et al. (2024): Deposit/Prospect, Showing or Occurrence. After merging and the data-coverage filter (Section 2.6), the national label table holds 66 occurrences (Fig. 2a). Within the three study provinces, the same steps reduced 21, 13 and 51 pooled records to 15 (Superior), 12 (Grenville) and 27 (Cordillera) positives; the remaining 12 occurrences lie elsewhere in Canada.

## 2.4 Geophysical and structural evidence layers

Table 1 lists the evidence layers. Seventeen national rasters cover the trap and the deep-architecture components: Bouguer gravity and its derivatives and filtered versions, total magnetic intensity and its differential reduction to the pole with derivatives (Jobin et al., 2017; Miles & Oneschuk, 2016), together with long-wavelength gravity and magnetics, LAB and Moho depth from the Tri-National Critical Minerals Mapping Initiative release (McCafferty et al., 2023). The Bouguer anomaly (density 2.67 g cm$^{-3}$, terrain-corrected), its first vertical and total horizontal derivatives and the total magnetic intensity are the public Natural Resources Canada compilations (Jobin et al., 2017; Miles & Oneschuk, 2016), regridded to 2 km (EPSG:3978) and 0.01°. The other seven gravity and magnetic grids were computed from them for this study. The 40 and 100 km high-pass versions of the Bouguer anomaly apply a sixth-order Butterworth filter, $H(k) = [1 + (k_0/k)^6]^{-1}$ with $k_0$ the inverse of the cut-off wavelength, in the wavenumber domain. The magnetic field was reduced to the pole differentially (Arkani-Hamed, 1988; Cooper & Cowan, 2005a): the total magnetic intensity was processed in 8° tiles with 4° overlap, each projected to a 1 km azimuthal equidistant grid, the reduction to the pole for induced magnetisation (Blakely, 1995) was computed for nine combinations of inclination and declination spanning the tile and interpolated to the field direction at each node, taken from the International Geomagnetic Reference Field (IGRF-13, epoch 2020.0; Alken et al., 2021), and the tiles were blended with cosine tapers. First vertical derivatives were computed in the wavenumber domain, total horizontal derivatives (THD) by central differences on metric cell sizes, and the tilt angle of a grid as the arctangent of its vertical over its total horizontal derivative (Miller & Singh, 1994; Verduzco et al., 2004); the THD tilt layers are the tilt angle of the

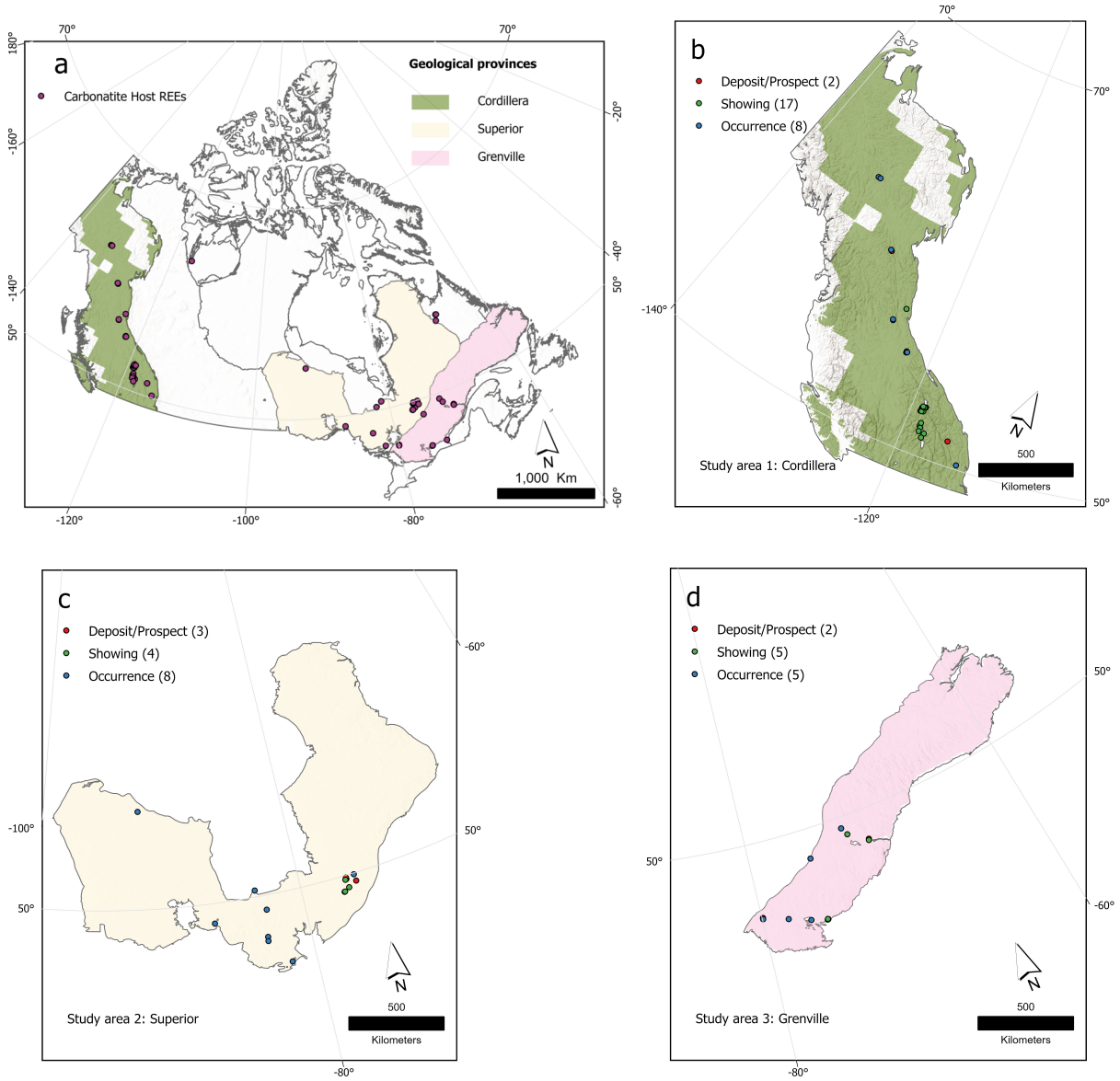



**Fig. 2** Comparison between the entire territory of Canada (a) and three priority areas (b-d) and known REE occurrences, illustrating the highly uneven and geologically constrained spatial distribution of REE mineralization across Canada. The coloured areas denote the regions covered by available data (the sampleable domain), within which sampling and prediction were confined. Occurrences compiled from two sources: Parsa et al. (2024) and Orris et al. (2018) Symbols denote the development status of each occurrence following Parsa et al. (2024): Deposit/Prospect (promising to well-defined mineralisation with confirmed or potentially economic resources, requiring further detailed exploration to delineate), Showing (a higher likelihood of economic interest but requiring more exploration), and Occurrence (the most basic indication of minerals, often without economic viability). Only occurrences retained after deduplication and domain filtering were used as positive samples.

THD grids. Two line layers (ancient passive margins, D.C. Bradley, 2008; faults) and four point layers (shallow and deep gravity and magnetic worms; McCafferty et al., 2023) enter as normalised proximities. GLCM contrast, homogeneity, energy and correlation were computed in 9×9-pixel windows on all eleven gravity and magnetic grids; feature pruning (Section 3.2) removes the redundant ones. Following Luo et al. (2026), the mean and standard deviation of the Bouguer anomaly, its 40 km high-pass, the reduced-to-pole magnetics and the total magnetic intensity within 10 km of each point were added as neighbourhood statistics. Radiometric grids (K, eU, eTh) were considered but not used, because their coverage of the study areas is incomplete and uneven; whole-rock geochemistry is too sparse to give a

value at every map cell. The province polygons of Wheeler et al. (1996) define the study areas only; no distance-to-province-boundary feature was used.

### 2.5 Geology

Geology enters from a hexagonal H3 grid (resolution 7, cells of about 5 km$^2$) attributed from the Geological Map of Canada (Wheeler et al., 1996). Each cell carries a major lithology class (for example Igneous Intrusive Alkalic or Metamorphic Gneiss Paragneiss), its dominant era, and flags for cells crossed by a lithological or an age (period) contact; each point takes the attributes of the cell that contains it. The lithology class enters as one-hot columns (a class gets its own column if at least 10 of the region's 5,000 training-background points fall in it; rarer classes share one), together with the era one-hot and the two contact flags. Following Luo et al. (2026), geology also enters as distances (km) to six elements of a generalised map, from which every lithological unit smaller than 20 connected cells (about 100 km$^2$) and the alkalic-intrusion class were removed: lithological contacts, age contacts, greenstone (extrusive) units, mafic–ultramafic intrusions, high-grade metamorphic units and sedimentary units. After pruning (Section 3.2) the inputs number 57 in Superior (33 geophysical and structural layers, 14 of them GLCM textures, 3 neighbourhood statistics, 2 regional distances, 15 lithology columns, 2 era columns and 2 contact flags), 61 in Cordillera and 66 in Grenville.

At the scale of a carbonatite complex a bedrock map can mark the target itself: in Superior, 6 of the 15 occurrences fall on cells mapped as alkalic intrusive rock, against 16 of the 3,000 evaluation-background points (0.5%). A mapped alkaline complex is legitimate evidence for an explorer, but a model can score well simply by looking it up. We therefore also report the models without the point-level map classes (the lithology, era and contact columns), keeping the geophysical and structural layers, neighbourhood statistics and regional distances (38, 37 and 41 inputs after pruning). This measures what the models achieve on ground where the map does not show a complex.

### 2.6 Sampleable domain and background

Rasters were clipped to each study-area outline buffered outward by one GLCM window radius (8 km), so that windows near the boundary read real data, and sampling and prediction were restricted to the domain where every layer holds data and every texture window is computable. This domain covers 98.6% of the Superior, 97.2% of the Grenville and 73.5% of the Cordilleran outline and 8.09 million km$^2$ (65.6%) of Canada. Background candidates are the nodes of a 0.1° grid inside the domain and more than 10 km from any known occurrence (19,371 in Superior). From these, two disjoint pools were drawn once per region at random: an evaluation background of 3,000 points and a training background of 5,000 points. Every model in this study is trained on the positives of its training fold plus the same 5,000 training-background points and scored on the held-out positives plus the same evaluation background. The prevalence in the scored pool is therefore 15/3,015 $\approx$ 0.5% in Superior, which is the AP of a random ranking.

For the national analyses the same rules were applied to Canada as a whole. Inside the study provinces the national background points are provincial ones (in Cordillera and Grenville random subsets of them, so that the density matches Superior's), and elsewhere background was drawn at Superior's density, 41% of the candidate nodes, with evaluation and training background split 3:5 as in each province. The national table holds 66 occurrences, 20,855 evaluation-background and 34,761 training-background points (prevalence 0.32%).

## 3 Methods

Figure 3 summarises the workflow. Code for every analysis, including the exact numbers in every table, is available in the repository listed under Code availability.

### 3.1 National and provincial inputs

Three inputs depend on the extent of the area over which they are computed: raster values at the edge of a clip (a coarse pixel whose centre falls outside the buffered outline is nodata), the GLCM grey levels, which are quantised between the 2nd and 98th percentiles of the clipped raster, and the generalised geology map, which is built from the H3 cells in the outline's bounding box plus a margin. All three

**Table 1** Evidence layers by mineral-system component and mappable criterion, with their geometry, native resolution and source. Rasters were sampled at points in their native projection and resolution; line and point layers enter as normalised proximities.

| Component | Mappable criterion | Layer | Geometry | Native resolution | Description | Source |
|---|---|---|---|---|---|---|
| Source | Mantle melting at craton margins | Ancient passive margins | Line | – | Proximity to ancient passive margins | D.C. Bradley (2008) |
| | Lithospheric architecture | LAB depth | Raster | 0.1° (≈11.1 km) | Depth to the lithosphere–asthenosphere boundary | McCafferty et al. (2023) |
| | Deep crustal architecture | BA upward-continued 30 km | Raster | 0.031° (≈3.4 km) | Bouguer gravity upward-continued by 30 km | McCafferty et al. (2023) |
| | | BA upward-continued 30 km HGM | Raster | 0.031° (≈3.4 km) | Horizontal-gradient magnitude of the upward-continued gravity | McCafferty et al. (2023) |
| | | Deep-source RTP | Raster | 0.018° (≈2.0 km) | Long-wavelength (deep-source) reduced-to-pole magnetics | McCafferty et al. (2023) |
| | | Deep-source RTP HGM | Raster | 0.019° (≈2.1 km) | Horizontal-gradient magnitude of the deep-source RTP | McCafferty et al. (2023) |
| Transport | Crustal architecture | Moho depth | Raster | 0.25° (≈27.8 km) | Depth to the Moho | McCafferty et al. (2023) |
| | | Faults | Line | – | Proximity to faults | McCafferty et al. (2023) |
| | | Shallow gravity worms | Point | – | Proximity to horizontal-gradient maxima of the Bouguer gravity | McCafferty et al. (2023) |
| | | Deep gravity worms | Point | – | Proximity to horizontal-gradient maxima of the 30 km upward-continued gravity | McCafferty et al. (2023) |
| | | Shallow magnetic worms | Point | – | Proximity to horizontal-gradient maxima of shallow magnetic sources | McCafferty et al. (2023) |
| | | Deep magnetic worms | Point | – | Proximity to horizontal-gradient maxima of deep magnetic sources | McCafferty et al. (2023) |
| Trap | Gravity anomaly | Bouguer anomaly (BA) | Raster | 2 km | Bouguer anomaly (2.67 g cm$^{-3}$, terrain-corrected) | Jobin et al. (2017) |
| | | BA 1VD | Raster | 2 km | First vertical derivative of the BA | Jobin et al. (2017) |
| | | BA THD | Raster | 2 km | Total horizontal derivative of the BA | Jobin et al. (2017) |
| | | BA THD tilt | Raster | 2 km | Tilt angle of the BA THD | This study |
| | | BA high-pass 40 km | Raster | 2 km | 40 km high-pass (sixth-order Butterworth) of the BA | This study |
| | | BA high-pass 100 km | Raster | 2 km | 100 km high-pass (sixth-order Butterworth) of the BA | This study |
| | Magnetic anomaly | Total magnetic intensity (TMI) | Raster | 0.01° (≈1.1 km) | Total magnetic intensity | Miles and Oneschuk (2016) |
| | | DRTP | Raster | 0.01° (≈1.1 km) | Differential reduction to the pole of the TMI | This study |
| | | DRTP 1VD | Raster | 0.01° (≈1.1 km) | First vertical derivative of the DRTP | This study |
| | | DRTP THD | Raster | 0.01° (≈1.1 km) | Total horizontal derivative of the DRTP | This study |
| | | DRTP THD tilt | Raster | 0.01° (≈1.1 km) | Tilt angle of the DRTP THD | This study |
| | Anomaly geometry | GLCM textures | Derived | 9×9-pixel window | Contrast, homogeneity, energy and correlation of each of the eleven gravity and magnetic grids | This study |
| | | Neighbourhood statistics | Derived | 10 km radius | Mean and standard deviation of the BA, BA high-pass 40 km, DRTP and TMI | This study, after Luo et al. (2026) |
| | Host architecture | Lithology, era, contacts | H3 grid | Resolution 7 (≈4.9 km$^2$ cells) | Major lithology class and era of the cell (one-hot); flags for lithological and age contacts in the cell | Wheeler et al. (1996) |
| | | Distances to generalised geology | Derived | – | Distances to lithological and age contacts and to greenstone, mafic–ultramafic, high-grade metamorphic and sedimentary units (Section 2.5) | This study, from Wheeler et al. (1996) |

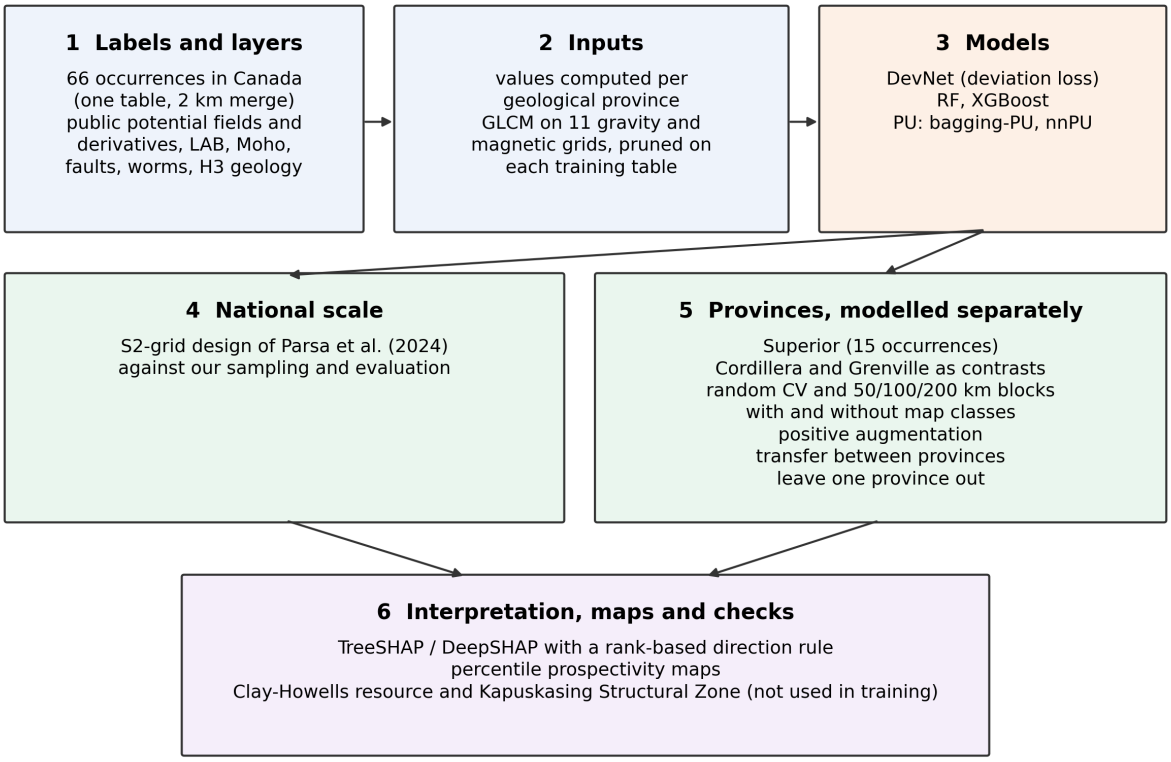



**Fig. 3** Workflow. The national experiments compare sampling designs; the provinces are modelled separately, and transfer between them is tested with the provincial models. Within a comparison, all models are trained on the same positives and background and scored on the same folds; only the treatment of the background differs.

were computed per geological province, with the provincial rasters and bounding boxes for the study provinces, so that every national value equals the provincial one at the same point. Feature pruning (Section 3.2) and the H3 class list are applied to the table on which a model is trained: national models use the 71 inputs pruned on the national table, and the model of a province uses that province's inputs and network capacity, also when it is applied to another province.

## 3.2 Feature pruning

The candidate features include many collinear geophysical derivatives. Within each region, features were clustered hierarchically on $1-|\rho_S|$ (Spearman correlation, average linkage) and cut at $|\rho_S| = 0.7$, keeping one representative per cluster chosen by lowest missing rate and then highest variance (Dormann et al., 2013). The rule never sees labels and therefore cannot leak information across folds.

## 3.3 Models

All five models are trained on the same rows, the positives $P$ of a training fold ($n_P$ points, $y = 1$) and the shared training background $U$ ($n_U = 5{,}000$ points, $y = 0$), and their scores are used only as rankings. They differ in what they assume about $U$: RF and XGBoost treat it as negative, bagging-PU and nnPU as an unlabelled mixture of negatives and undiscovered positives, and DevNet as a sample of ordinary ground from which occurrences deviate.

**Random Forest** (RF; Breiman, 2001) classifies $P$ against $U$ with fully grown trees. Class weights $w_1 = (n_P + n_U)/(2n_P)$ and $w_0 = (n_P + n_U)/(2n_U)$ balance the two classes, and the score is the mean over the $T$ trees of the weighted share of positives in the leaf that $\mathbf{x}$ reaches,

$$s_{\mathrm{RF}}(\mathbf{x}) = \frac{1}{T}\sum_{t=1}^{T} \frac{w_1\, n_{t,1}(\mathbf{x})}{w_1\, n_{t,1}(\mathbf{x}) + w_0\, n_{t,0}(\mathbf{x})}, \tag{1}$$

where $n_{t,1}(\mathbf{x})$ and $n_{t,0}(\mathbf{x})$ count the positive and background training points in that leaf. We use $T = 300$ trees with $\sqrt{p}$ of the $p$ inputs tried at each split, after median imputation and standardisation fitted on the training fold. With about 12 training positives against 5,000 background points most held-out points score exactly zero, so the seed is varied across folds and the resulting variance enters the reported intervals.

**XGBoost** (T. Chen & Guestrin, 2016) also labels $U$ as negative, but builds an additive ensemble $F(\mathbf{x}) = \sum_{k=1}^{K} f_k(\mathbf{x})$ of shallow regression trees by gradient boosting of the weighted logistic loss

$$\mathcal{L} = \sum_i v_i \, \ell\big(y_i, F(\mathbf{x}_i)\big) + \sum_{k=1}^{K} \Big(\gamma J_k + \tfrac{1}{2}\lambda \|\boldsymbol{\omega}_k\|_2^2 + \alpha \|\boldsymbol{\omega}_k\|_1\Big), \tag{2}$$

where $v_i = n_U/n_P$ for positives and 1 for background, and $J_k$ and $\boldsymbol{\omega}_k$ are the number of leaves and the leaf values of tree $k$. We use $K = 200$ trees of depth 2, learning rate 0.1, row and column subsampling of 0.8, minimum child weight 3, $\lambda = 2$, $\alpha = 0.5$ and $\gamma = 0$, on median-imputed inputs.

**Bagging-PU** (Mordelet & Vert, 2014) treats $U$ as unlabelled. Each of $B = 200$ fully grown decision trees $h_b$ ($\sqrt{p}$ inputs per split) is trained on all positives against a bootstrap sample $U_b$ of $n_P$ background points provisionally labelled 0, and the score is

$$s_{\mathrm{BPU}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} h_b(\mathbf{x}). \tag{3}$$

An undiscovered occurrence in $U$ is drawn as a negative by few if any of the trees, so averaging dilutes its effect. Inputs are median-imputed and standardised.

**nnPU** (Kiryo et al., 2017) also treats $U$ as unlabelled, but corrects the classification risk for the positives hidden in it, given the class prior $\pi = P(y = 1)$ of the unlabelled population. With $\hat{R}_P^{\pm}(g) = n_P^{-1} \sum_{\mathbf{x}\in P} \ell\big(g(\mathbf{x}), \pm 1\big)$ and $\hat{R}_U^{-}(g) = n_U^{-1} \sum_{\mathbf{x}\in U} \ell\big(g(\mathbf{x}), -1\big)$, it minimises the non-negative risk

$$\hat{R}(g) = \pi \, \hat{R}_P^{+}(g) + \max\big\{0, \; \hat{R}_U^{-}(g) - \pi \, \hat{R}_P^{-}(g)\big\} \tag{4}$$

with the sigmoid loss $\ell(z, t) = 1/(1 + e^{tz})$; whenever the bracketed term turns negative, a step that increases it is taken instead. We use $\pi = 0.01$ (sensitivity in Appendix A, Table A2) and exactly DevNet's network, inputs, optimiser, number of steps and five-restart ensemble (below), with all training positives and 256 background rows in each step, so that DevNet versus nnPU isolates the PU risk from the deviation loss.

**DevNet** (Pang et al., 2019) treats $U$ as a sample of ordinary ground rather than as negatives. A network maps each input vector $\mathbf{x}$ to a scalar score $\phi(\mathbf{x})$, whose deviation $d(\mathbf{x}) = (\phi(\mathbf{x}) - \mu_R)/\sigma_R$ is measured against the mean $\mu_R$ and standard deviation $\sigma_R$ of a fixed reference sample of 50,000 standard-normal draws, and the loss

$$L = (1 - y)\,|d(\mathbf{x})| + y \, \max\{0, \, a - d(\mathbf{x})\} \tag{5}$$

pulls background scores towards the reference and pushes positives beyond a margin $a = 5$, the value of Pang et al. (2019) (Appendix D, Table D7). Background points are thus never required to score low, only to look typical. The network has one hidden ReLU layer with dropout (64 units, dropout 0.3 and weight decay $10^{-4}$ in Cordillera; 32, 0.4 and $10^{-3}$ in Superior and Grenville) and is trained for 300 Adam steps (Kingma & Ba, 2014) with learning rate $2 \times 10^{-3}$, cosine annealing to $10^{-5}$ and gradient norm clipped at 1; each fold score is the rank average of five restarts with different seeds. Inputs are mapped to standard-normal marginals by a quantile transformation (1,000 quantiles) fitted on the training fold, because many evidence layers are strongly skewed (inverted distances concentrated near 0 or 1, long-tailed textures, 0/1 dummies) and, unlike tree ensembles, multilayer perceptrons are sensitive to such marginal distributions (Gorishniy et al., 2021; Grinsztajn et al., 2022). Each step uses 16 positives drawn with replacement and 240 background rows, so that over 300 steps every background row is seen about 14 times, against about once with label-balanced batches of 16 + 16 rows; Appendix D tests both choices.

The calibration approach of Elkan and Noto (2008) is not included: under its assumptions the calibrated output is a monotone transform of the positive-versus-background classifier, so its ranking, and every rank metric reported here, equals RF's.

### 3.4 Evaluation protocol

We emphasise average precision (AP) because, at 0.5% prevalence, it reflects the practical task of concentrating occurrences into a small search area, whereas the area under the ROC curve (AUC) is dominated by the bulk of easily rejected background (Saito & Rehmsmeier, 2015). Exploration efficiency is also shown as capture-rate curves: the share of held-out occurrences ranked within a given top fraction of the scored pool (Agterberg & Bonham-Carter, 2005; Chung & Fabbri, 2003). Because the evaluation background is a random sample of the domain, this fraction approximates search area. Because unexplored background may host undiscovered occurrences, all metrics are conservative.

**Random cross-validation.** Repeated stratified 5-fold cross-validation with six repeats is used, with identical splits for all models. With three or four positives per validation fold, a per-fold AUC is unstable, and undefined for a fold without positives, so averaging over folds is not workable here. Out-of-fold scores are therefore pooled (A.P. Bradley, 1997): each held-out point is scored once per repeat by the model of its fold, the scores are converted to within-fold percentile ranks so that the folds are comparable, and one AUC and one AP are computed over all folds of the repeat. Pooling compares positives and background scored by different fold models, which is known to bias AUC downwards (Airola et al., 2019; Parker et al., 2007), so the pooled values are conservative rather than optimistic. The six repeats give six pooled values per model, reported as the median and the 2.5–97.5 percentile interval (labelled 95% interval).

**Spatially blocked cross-validation.** Random folds can hold out an occurrence a few kilometres from a training occurrence. Following Roberts et al. (2017) and Valavi et al. (2019), labelled points were assigned to square blocks of 50, 100 or 200 km side in Canada Atlas Lambert (EPSG:3978), whole blocks were held out together using stratified group 5-fold splitting (fewer folds if fewer blocks contain positives), and every training-background point falling inside a held-out block was removed from training for that fold. Each of six repeats uses a random offset of the block grid, and the out-of-fold scores are pooled as above.

Because all models share every split, each repeat compares them on identical data. We report in how many repeats one model has the higher AP and, for the difference in AP, a 95% percentile interval from 5,000 stratified bootstrap resamples of positives and background, with each point's out-of-fold rank averaged over the repeats (Efron & Tibshirani, 1993). With 12–27 positives these comparisons have little power, so a difference is treated as established only when it is consistent across repeats, block sizes and provinces.

### 3.5 National-scale experiments

**Sampling and augmentation.** We rebuilt the national S2-grid design of Parsa et al. (2024) with the same geometry library on every level-12 S2 cell (about 4.5 $km^2$) whose centre lies in the national domain (1,616,693 cells), with the inputs extracted at the cell centres; the 66 occurrences fall in 66 cells. As in the original study, each occurrence cell was replaced by its four level-13 children, an equal number of negatives was drawn from the 5% of cells with the lowest mean cosine similarity to the positives in min–max scaled feature space, and both classifiers of the original study were evaluated on min–max scaled inputs by test AUC over 50 stratified 70/30 splits of the augmented labels: a random forest of 1,000 trees, within the range of the original grid search (which we did not repeat), and the best model of that study, a one-dimensional CNN with one convolutional layer (512 filters of width 5, with batch normalisation and dropout), max pooling, dense layers of 256, 128 and 64 units and a sigmoid output, trained with Adam on a mean-squared-error loss with early stopping. The CNN settings that study does not report are ours: pool size 2, dropout 0.3, batch size 32, learning rate $10^{-3}$, and early stopping after 20 epochs without improvement on a stratified 20% of the training labels. In our design the same two models, class-balanced (balanced class weights for the forest, class-balanced resampling in every epoch for the CNN), are trained on one label per occurrence cell against a random sample of 34,761 cells more than 10 km from any occurrence, and tested on the held-out 30% of occurrence cells against a random 30% of a disjoint sample of 20,855 cells, a realistic prevalence of about 0.3%. To see which element of the original design drives its score, we also changed one element at a time: random instead of cosine-dissimilar negatives, a split that keeps the four children of a cell on the same side, and no augmentation.

For the maps, each design and classifier was fitted once to all of its labels and scored every L12 cell, and the share of 50,000 random cells whose nearest training row in the scaled inputs is a positive describes what each design teaches its models.

### 3.6 Positive augmentation

To test whether positive augmentation can compensate for label scarcity, each training fold's real positives were supplemented with $m \times n_{\text{pos}}$ augmented positives ($m = 1, 3, 10$) before RF and DevNet were retrained; the held-out positives were always real. Three generators were compared, all fitted in the standardised feature space of the training fold only: Gaussian jitter of real positives ($\sigma = 0.1$ standard deviations); SMOTE-style interpolation between a positive and one of its three nearest positive neighbours (Chawla et al., 2002); and a tabular Wasserstein GAN with gradient penalty (WGAN-GP; Arjovsky et al., 2017; Goodfellow et al., 2014; Gulrajani et al., 2017) with an eight-dimensional latent input, two hidden layers of 64 units in generator and critic, five critic updates per generator update, gradient-penalty weight 10 and 2,000 generator updates, trained on the fold's 12 positives. The geology columns of augmented positives were snapped back to valid one-hot and 0/1 values. To see what the generators produce, each was also fitted to all 15 Superior positives and 150 samples were characterised by the distance to the nearest real positive divided by the median nearest-neighbour distance among real positives (a value near 0 indicates copying), by the share of samples whose nearest labelled neighbour is a positive rather than a background point, and by their DevNet score percentile.

### 3.7 Transfer between provinces

The model of province A (its inputs, capacity, occurrences and training background) scored the occurrences and evaluation background of province B, with six seeds; the diagonal, random cross-validation within A, is the provincial result. In leave-one-province-out validation a model trained on the national table without one province group (Superior, Cordillera, Grenville or the rest of Canada) scored that group. To describe the differences between provinces, a random forest (300 trees, 5-fold cross-validation) tried to tell the training background of two province groups apart, and the occurrences of two provinces; and the AUC of each input alone, for a province's occurrences against its evaluation background, gives the direction in which the input marks occurrences there.

### 3.8 Interpretation and mapping

Production models were fitted to all positives and the training background of each region. SHAP values (Lundberg & Lee, 2017) were computed with TreeSHAP for RF (Lundberg et al., 2020) and DeepSHAP for DevNet on an explain set of all positives plus evaluation-background points (200 in total), with 100 training-background points as DevNet's reference sample; DevNet's SHAP values refer to its quantile-transformed inputs, which leaves the rank-based directions defined below unaffected. DevNet SHAP values were averaged over the five ensemble members. Importance is mean |SHAP|. The *direction* of a feature's effect is the Spearman correlation between the feature's value and its SHAP value over the explain set (for one-hot dummies, the sign of the mean SHAP difference between presence and absence), with $|\rho| < 0.1$ reported as no clear direction. We do not use the sign of the mean SHAP value, which reflects where the explained points lie relative to the explainer's baseline rather than whether higher feature values raise the score: for RF with balanced class weights the TreeSHAP baseline is near 0.5 while nearly all explained points score near 0, so that rule would label every RF feature as "higher → less prospective". Prospectivity maps classify each model's scores on a 0.1° prediction grid into the top 1%, 5% and 10%, because DevNet's anomaly score and RF's probability are not on a common scale. As an external check, the production models scored Clay-Howells, a drill-tested Fe–REE–Nb prospect with reported resources on the KSZ (Ontario Geological Survey, 2011) that is not among the training labels, at its recorded location.

## 4 Results

### 4.1 National scale

Rebuilt on our national data, the design of Parsa et al. (2024) gives a test AUC of 1.000 with both its random forest and its CNN (Table 2), as high as the published 0.91–0.96 or higher. With one label per

**Table 2** The national S2-grid design of Parsa et al. (2024) and ours, with the random forest (1,000 trees) and the one-dimensional CNN of the original study on min–max inputs: mean test AUC and AP over 50 stratified 70/30 splits. Test set: occurrence share among the test points.

| | | | | | Random forest | | CNN | |
|---|---|---|---|---|---|---|---|---|
| Design | Positive labels | Negatives | Split | Test set | AUC | AP | AUC | AP |
| Parsa et al. (2024) | four L13 children per occurrence cell (264) | 264 cells from the 5% least cosine-similar | child labels, random 70/30 | balanced (50%) | 1.000 | 1.000 | 1.000 | 1.000 |
| This study | one per occurrence cell (66) | 34,761 random cells >10 km from occurrences | occurrence cells, random 70/30 | realistic (0.3%) | 0.903 | 0.292 | 0.852 | 0.059 |

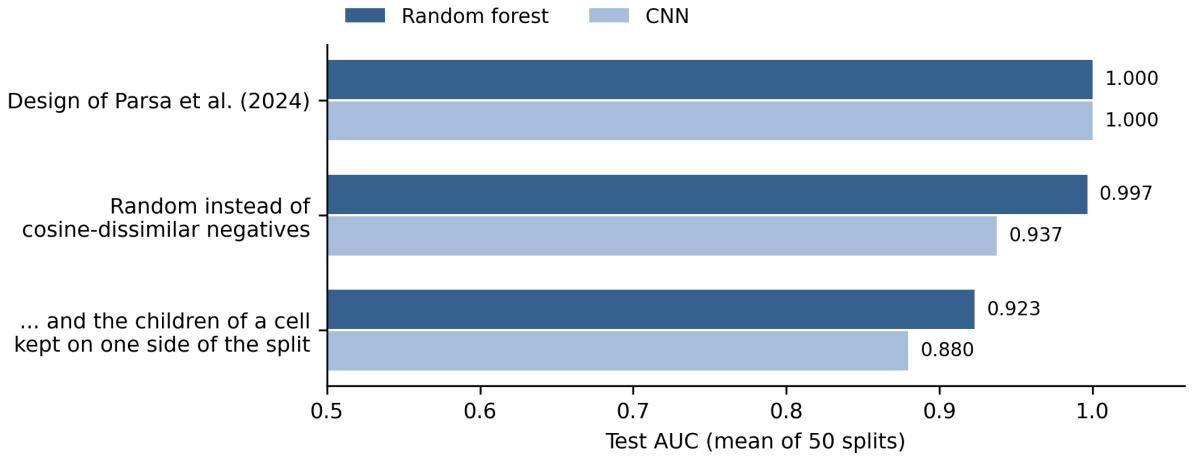



**Fig. 4** Test AUC of the random forest and the CNN of Parsa et al. (2024) (mean over 50 stratified 70/30 splits of a balanced test set) in their S2-grid design, with random instead of cosine-dissimilar negatives, and with the four children of an occurrence cell also kept on the same side of the split, so that no test positive has a sibling in training.

occurrence cell, random background and a realistic prevalence, the forest reaches an AUC of 0.903 and an AP of 0.292, and the CNN an AUC of 0.852 and an AP of 0.059. Negatives drawn from the ground least similar to the occurrences suffice on their own for a near-perfect score, because that ground is easy to reject: without any augmentation they give 0.998 (forest) and 0.996 (CNN). Splitting augmented children at random leaves a sibling of 97% of the test positives in the training set; with random negatives this gives the forest 0.997 and the CNN 0.937, against 0.923 and 0.880 when the four children of a cell are kept on the same side (Fig. 4). Cosine-dissimilar negatives also make worse models: trained against them rather than against random negatives in an otherwise identical design, the AP at realistic prevalence falls from 0.190 to 0.113 for the forest and from 0.025 to 0.017 for the CNN.

The two designs also produce very different maps (Fig. 5). In the design of Parsa et al. (2024) the forest assigns a probability of 0.5 or more to 62.5% of the cells and the CNN to 57.5%. Their negatives come from the 5% of cells least similar to the occurrences, a median 480 km from the nearest occurrence against 355 km for all cells, and so cover the margins of the input space (Fig. 6): whatever does not resemble those margins is grouped with the occurrences, and for 61.6% of 50,000 random cells the nearest training row is a positive. Against 34,761 random background cells the same forest gives 0.5 or more to 91 cells (less than 0.1%), 66 of them the occurrence cells themselves and the rest within 31 km of an occurrence, and 64.4% of the cells receive no positive vote from any of its 1,000 trees. The background now spans the input space, so the nearest training row is a positive for only 0.2% of the random cells, and a fully grown tree votes for an occurrence only where a cell closely resembles one; the 2.8% of cells at 0.01 or more lie a median 65 km from an occurrence, 79% of them in the Cordillera, Grenville and Superior. The CNN, with the settings of Parsa et al. (2024) and our labels and background, is not selective: 36.9%

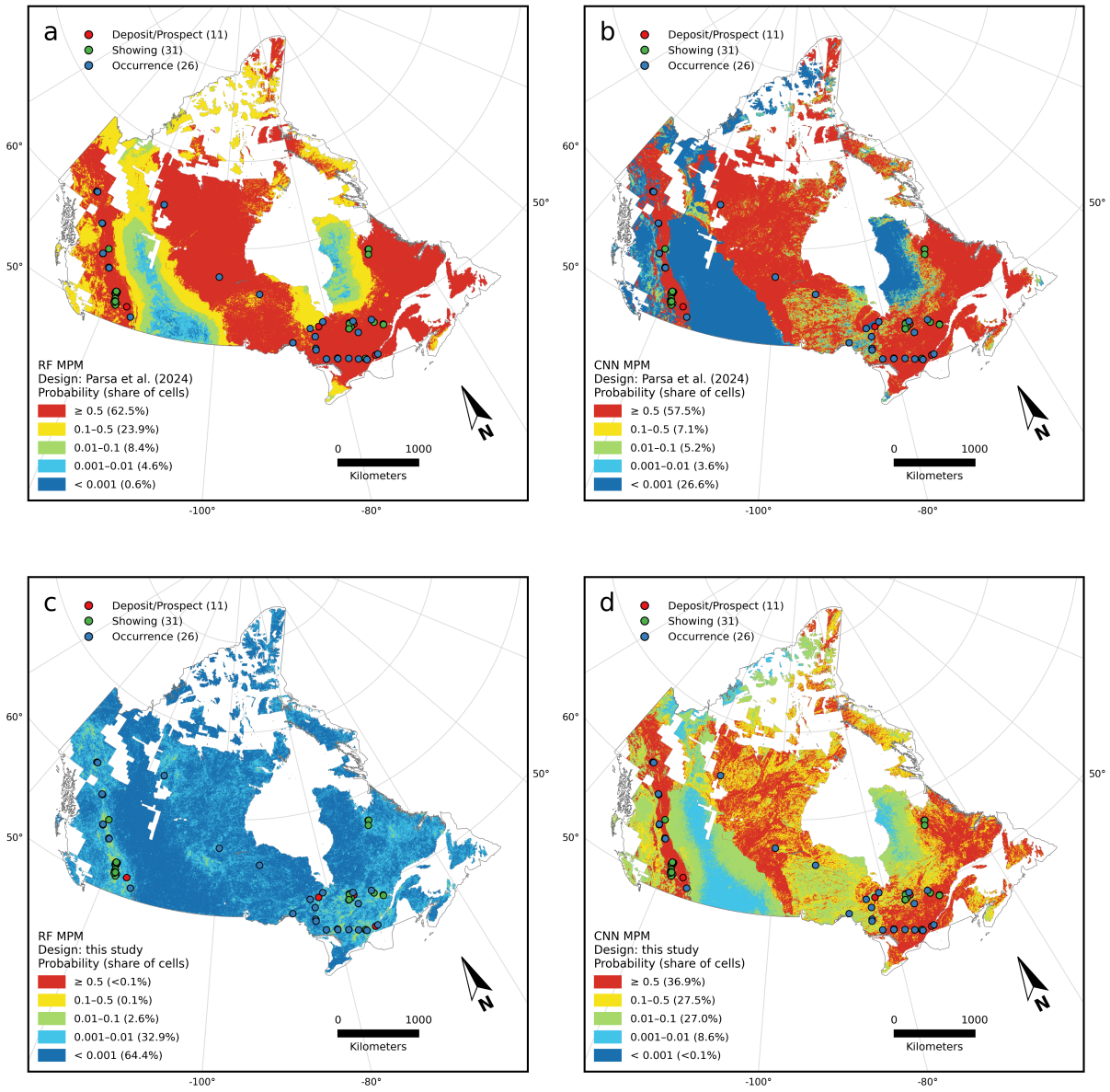



**Fig. 5** National prospectivity of the 1,616,693 S2 level-12 cells under the two designs of Table 2, each model fitted to all labels of its design: (a) random forest and (b) CNN in the design of Parsa et al. (2024); (c) random forest and (d) CNN in ours. The classes of predicted probability are common to the four maps (share of the cells in brackets); occurrences by development class (Canada Atlas Lambert). White: outside the national domain.

of the cells reach 0.5 and 10.7% reach 0.9, against 48.6% at 0.9 in their design. Its map singles out no part of the country and, like its ranking of held-out occurrences (AP 0.059), carries little information for targeting. Neither design gives calibrated probabilities, since both train at a balanced class ratio, so the maps compare designs rather than probabilities of mineralisation. With the same forest and inputs, however, the choice of negatives alone moves the high-probability area from most of Canada to the immediate surroundings of the known occurrences. The forest's probabilities are low at the provincial scale too: in Superior the production forest gives a probability of exactly zero to 80% of the prediction grid (Section 4.2.7), and its maps are to be read as rankings rather than as probabilities.

The national comparison concerns how labels and negatives are sampled; it does not show that one model should serve the whole country. Canadian carbonatites were emplaced in several episodes from about 2.68 Ga onwards (Rukhlov & Bell, 2010) and in different settings, from the Archean Superior craton to the Cordilleran Orogen, the Grenville orogen and the St. Lawrence rift system (Groulier et al., 2020; Néron et al., 2018), so the same evidence layer need not stand for the same process everywhere: a

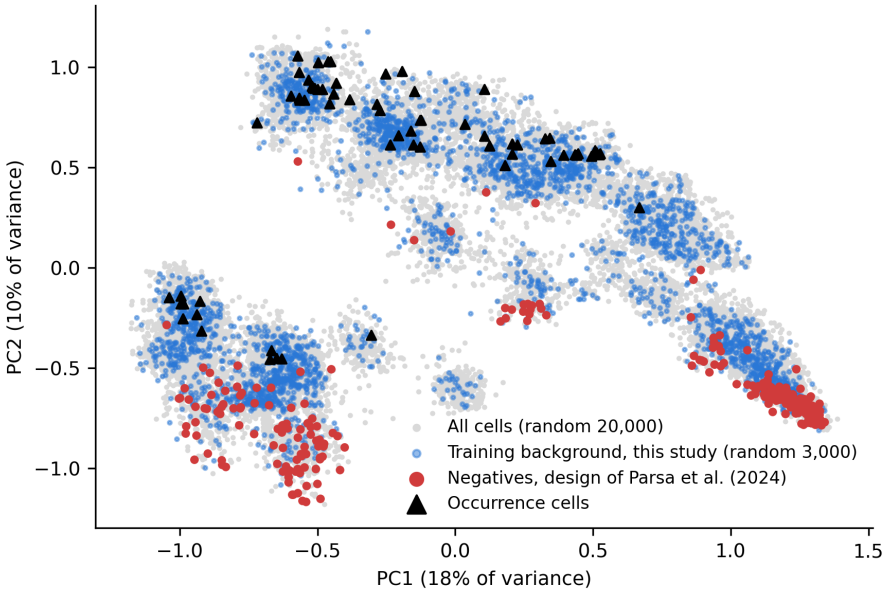



**Fig. 6** Where the negatives of the two designs lie in the inputs: the first two principal components of the min–max scaled national inputs (fitted on 50,000 random cells), with the 66 occurrence cells, the 264 negatives of the design of Parsa et al. (2024), drawn from the 5% of cells least cosine-similar to the occurrences, and a random 3,000 of the 34,761 cells of our training background.

mappable criterion is only as useful as it is representative of the process it stands for, and its significance changes with the scale at which the critical processes operate (McCuaig et al., 2010). A model is, moreover, reliable only where its inputs resemble those it was trained on (Meyer & Pebesma, 2021). We therefore model the Superior, Cordilleran and Grenville occurrences separately (Section 4.2) and use the provincial models to test whether a model trained in one province applies to another (Section 4.2.5).

## 4.2 Provincial models

### 4.2.1 Model comparison under random cross-validation

In Superior all models except nnPU rank held-out occurrences far above chance (Table 3): AP is 0.20–0.38 against a random-ranking value of 0.005, and AUC is 0.97–0.99. DevNet has the highest median AP (0.381) and RF the highest AUC (0.985). DevNet's AP exceeds RF's, XGBoost's (0.265), bagging-PU's and nnPU's in all six repeated splits, by 0.18–0.30 on repeat-averaged ranks. nnPU, which shares DevNet's network and inputs, reaches an AP of only 0.159 and an AUC of 0.842.

The contrasting regions behave differently. In Cordillera, bagging-PU and RF lead (AP 0.469 and 0.456), and DevNet (0.327; lower than RF in all six repeats) and XGBoost (0.389) trail. In Grenville all models are weak; XGBoost leads (0.368), followed by RF (0.268), both ahead of DevNet (0.127) in all six repeats. nnPU fails in both provinces (AUC 0.58–0.61). That Superior is the best-predicted of the three provinces agrees with the province-level success rates of Parsa et al. (2024).

### 4.2.2 Spatially blocked cross-validation

Holding out whole blocks widens the gap between DevNet and the tree-based and PU learners in Superior (Fig. 7; Table 4). DevNet's AP is 0.371, 0.345 and 0.221 with 50, 100 and 200 km blocks. RF's falls from 0.268 under random splits to 0.165, 0.129 and 0.087, XGBoost's from 0.265 to 0.177, 0.119 and 0.098, and bagging-PU's to 0.133, 0.076 and 0.072. DevNet has the higher AP than RF, XGBoost and bagging-PU in all six repeats at every block size, and than nnPU in 16 of the 18, by +0.23 to +0.31 against RF, +0.21 to +0.33 against XGBoost, +0.25 to +0.39 against bagging-PU and +0.13 to +0.30 against nnPU. With 15 occurrences, however, the 95% intervals of the differences from RF reach down to 0.03–0.06, so the margin over RF is consistent but its size is uncertain. AUC differs less between models (Appendix A, Table A1): RF has the highest AUC with 50 km blocks (0.976 against 0.969 for DevNet), and DevNet

**Table 3** Pooled out-of-fold AUC and AP (median and 95% interval over six repeated 5-fold random splits). Bold: best of the five models in each region.

| Region ($n_{pos}$) | Model | AUC | AP |
|---|---|---|---|
| Superior (15) | DevNet | 0.972 | **0.381** [0.347, 0.452] |
| | RF | **0.985** | 0.268 [0.202, 0.314] |
| | XGBoost | 0.976 | 0.265 [0.243, 0.298] |
| | Bagging-PU | 0.976 | 0.198 [0.163, 0.219] |
| | nnPU | 0.842 | 0.159 [0.096, 0.210] |
| Cordillera (27) | DevNet | 0.925 | 0.327 [0.263, 0.353] |
| | RF | 0.936 | 0.456 [0.428, 0.548] |
| | XGBoost | 0.926 | 0.389 [0.357, 0.448] |
| | Bagging-PU | **0.937** | **0.469** [0.356, 0.485] |
| | nnPU | 0.575 | 0.023 [0.017, 0.037] |
| Grenville (12) | DevNet | 0.826 | 0.127 [0.060, 0.168] |
| | RF | 0.836 | 0.268 [0.202, 0.320] |
| | XGBoost | 0.870 | **0.368** [0.216, 0.429] |
| | Bagging-PU | **0.909** | 0.151 [0.133, 0.197] |
| | nnPU | 0.609 | 0.009 [0.008, 0.010] |

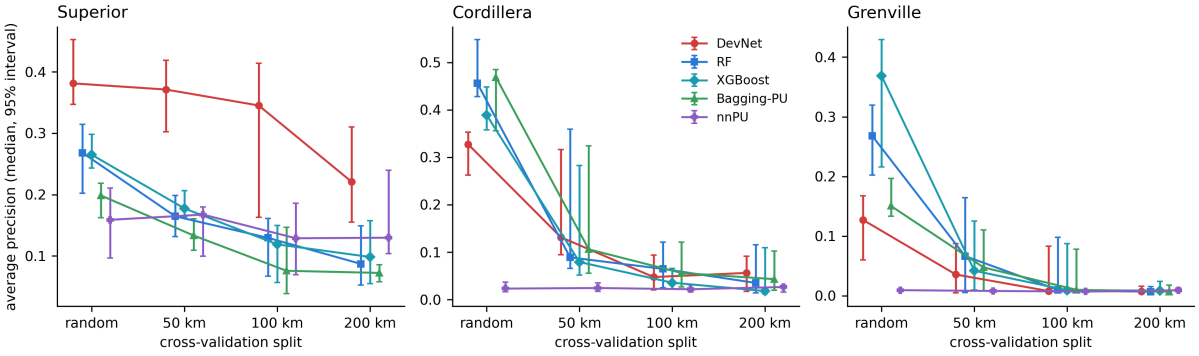



**Fig. 7** Average precision under random and spatially blocked cross-validation (median and 95% interval over six repeats). The random-ranking AP is 0.005 (Superior), 0.009 (Cordillera) and 0.004 (Grenville).

with 100 and 200 km blocks (0.928 and 0.886). Under 50 and 100 km blocks the occurrences that DevNet ranks lowest are Kapuskasing and Nemegosenda Lake, on the KSZ.

Capture-rate curves (Fig. 8) show where the AP differences arise. Under random splits DevNet and RF capture about half of the held-out occurrences in the top 1% of the ranked pool (53% and 50%), and by the top 10% RF captures all of them, bagging-PU 97% and DevNet 93%. With 50 and 100 km blocks DevNet keeps most of its top-1% capture (53% and 43%) and captures 93% and 87% by the top 10%, whereas RF captures 40% and 27% in the top 1% (100% and 77% by the top 10%) and bagging-PU 30% and 23% (90% and 70%). With 200 km blocks every model captures 40% or less in the top 1%, and by the top 10% DevNet and bagging-PU (70%) slightly exceed RF (63%).

In Cordillera every model degrades sharply under blocking (AP $\leq 0.13$ at 50 km and $\leq 0.07$ at 100–200 km), and in Grenville AP approaches the random-ranking value ($\leq 0.010$ at 100–200 km). In these regions no difference between models is of practical size: their random-split skill owes much to spatial proximity between training and test occurrences (Section 5.4).

### 4.2.3 Contribution of geological map classes

A single map class carries much of the skill in Superior. Removing the alkalic-intrusion column lowers DevNet's AP to 0.346, 0.260, 0.177 and 0.137 under random splits and 50, 100 and 200 km blocks, and RF's to 0.194, 0.126, 0.065 and 0.067: this one class, which covers 6 of the 15 Superior occurrences and 0.5% of the background, accounts for 30–49% of DevNet's spatially blocked AP. DevNet without it still has a higher AP than RF with it at every block size. Withholding all point-level map classes costs DevNet and RF about half or more of their AP: DevNet's falls to 0.159 under random splits and 0.123, 0.081 and 0.075 under blocks, RF's to 0.146 and 0.084, 0.044 and 0.050. Without map classes DevNet still has the

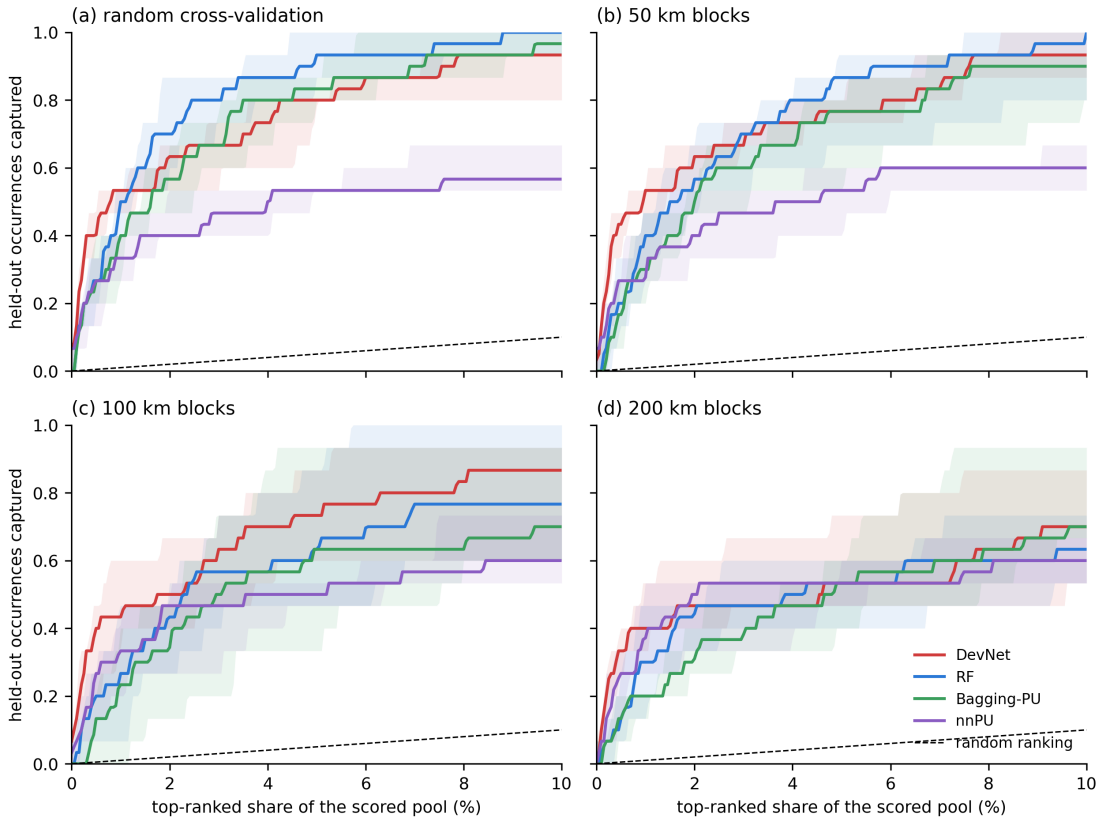



**Fig. 8** Superior capture-rate curves: share of held-out occurrences ranked within the top fraction of the scored pool (a proxy for search area) under (a) random cross-validation and (b) 50 km, (c) 100 km and (d) 200 km spatial blocks. Lines: median over six repeats; bands: range. Dashed: random ranking.

**Table 4** Average precision (median over six repeats) under random and spatially blocked cross-validation. Bold: highest AP in each row. AUC for the same runs is given in Appendix A (Table A1).

| Region | Split | DevNet | RF | XGBoost | Bagging-PU | nnPU |
|---|---|---|---|---|---|---|
| Superior | random | **0.381** | 0.268 | 0.265 | 0.198 | 0.159 |
| | 50 km | **0.371** | 0.165 | 0.177 | 0.133 | 0.167 |
| | 100 km | **0.345** | 0.129 | 0.119 | 0.076 | 0.129 |
| | 200 km | **0.221** | 0.087 | 0.098 | 0.072 | 0.130 |
| Cordillera | random | 0.327 | 0.456 | 0.389 | **0.469** | 0.023 |
| | 50 km | **0.131** | 0.089 | 0.079 | 0.106 | 0.024 |
| | 100 km | 0.048 | **0.065** | 0.036 | 0.056 | 0.022 |
| | 200 km | **0.056** | 0.035 | 0.018 | 0.043 | 0.027 |
| Grenville | random | 0.127 | 0.268 | **0.368** | 0.151 | 0.009 |
| | 50 km | 0.036 | **0.067** | 0.042 | 0.047 | 0.008 |
| | 100 km | 0.008 | 0.010 | 0.009 | **0.010** | 0.007 |
| | 200 km | 0.007 | 0.007 | 0.009 | 0.007 | **0.009** |

highest AP under every block size, ahead of XGBoost (0.105, 0.050 and 0.062) and bagging-PU (0.102, 0.058 and 0.066), while XGBoost and bagging-PU lead under random splits (0.175 and 0.174) and nnPU fails (AP 0.009).

### 4.2.4 Positive augmentation

The generators behave very differently (Table 6; Appendix B, Fig. B1). Jitter and SMOTE samples lie close to the real positives (median nearest-neighbour ratio 0.11 and 0.24) and almost all have a real positive as their nearest labelled neighbour (100% and 99%): they are near-copies of the training

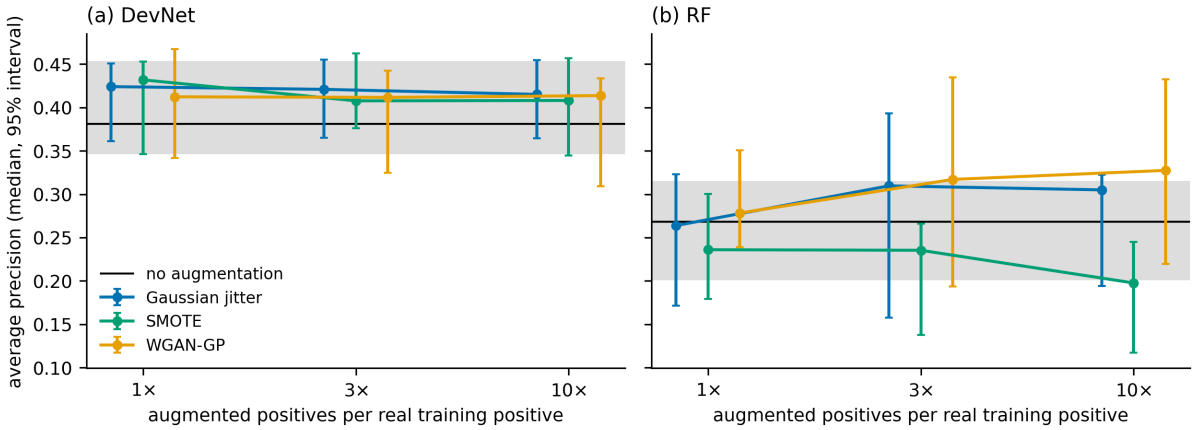



**Fig. 9** Superior average precision with fold-safe positive augmentation, by generator and number of augmented positives per real training positive. Line and grey band: no augmentation (median and 95% interval).

**Table 5** Fold-safe positive augmentation, Superior (random cross-validation, identical folds; held-out positives always real): AUC and AP (median and 95% interval over six repeats).

| Model | Generator | $m$ | AUC | AP |
|---|---|---|---|---|
| DevNet | none | – | 0.972 | 0.381 [0.347, 0.452] |
| | Gaussian jitter | 1× | 0.966 | 0.424 [0.361, 0.451] |
| | Gaussian jitter | 3× | 0.969 | 0.421 [0.365, 0.455] |
| | Gaussian jitter | 10× | 0.971 | 0.415 [0.364, 0.455] |
| | SMOTE | 1× | 0.973 | 0.432 [0.346, 0.453] |
| | SMOTE | 3× | 0.975 | 0.408 [0.376, 0.462] |
| | SMOTE | 10× | 0.970 | 0.408 [0.344, 0.457] |
| | WGAN-GP | 1× | 0.973 | 0.412 [0.342, 0.467] |
| | WGAN-GP | 3× | 0.968 | 0.412 [0.325, 0.442] |
| | WGAN-GP | 10× | 0.960 | 0.414 [0.309, 0.433] |
| RF | none | – | 0.985 | 0.268 [0.202, 0.314] |
| | Gaussian jitter | 1× | 0.964 | 0.264 [0.171, 0.323] |
| | Gaussian jitter | 3× | 0.983 | 0.310 [0.158, 0.393] |
| | Gaussian jitter | 10× | 0.980 | 0.305 [0.194, 0.323] |
| | SMOTE | 1× | 0.981 | 0.236 [0.179, 0.300] |
| | SMOTE | 3× | 0.983 | 0.235 [0.138, 0.266] |
| | SMOTE | 10× | 0.980 | 0.198 [0.117, 0.245] |
| | WGAN-GP | 1× | 0.984 | 0.278 [0.239, 0.351] |
| | WGAN-GP | 3× | 0.983 | 0.317 [0.193, 0.435] |
| | WGAN-GP | 10× | 0.984 | 0.327 [0.219, 0.433] |

occurrences. WGAN-GP samples are as far from the real positives as the real positives are from one another (ratio 0.97), but for 37% of them the nearest labelled point is a background point: with 15 training points the generator spreads probability mass into background-like parts of feature space. All three sets of samples receive high DevNet scores (median percentile 0.98).

The folds are identical with and without augmentation (Table 5; Fig. 9). For DevNet every generator raises the median AP somewhat (with augmentation its AP lies between 0.408 and 0.432, against 0.381 without), but not consistently: augmented DevNet has the higher AP in only two to five of the six repeated splits. For RF, jitter changes the AP from 0.268 to 0.264–0.310, SMOTE lowers it (0.198–0.236) and WGAN-GP raises it (0.278–0.327). The best augmented RF (WGAN-GP 10×, AP 0.327) remains below DevNet without augmentation (0.381). Under spatial blocking (Appendix B, Table B3) WGAN-GP raises RF's AP with 100 km blocks from 0.129 to 0.135–0.191 and with 200 km blocks from 0.087 to 0.122–0.132, still well below unaugmented DevNet (0.345 and 0.221), whereas jitter and SMOTE change it little (0.110–0.139 and 0.077–0.096). DevNet's AP changes little with 100 km blocks (0.308–0.341 against 0.343) and varies between 0.200 and 0.258 with 200 km blocks (against 0.216).

**Table 6** What the generators produce (fitted to all 15 Superior positives; 150 samples each). NN ratio: distance to the nearest real positive divided by the median nearest-neighbour distance among real positives (0 = copy). Nearest labelled neighbour: share of samples whose nearest point among positives and training background is a positive. DevNet percentile: median score percentile among the training background under the production DevNet.

| Source | NN ratio, median | NN ratio, P10 | NN is positive | DevNet pct. |
|---|---|---|---|---|
| Real occurrences (leave-one-out) | 1.00 | 0.60 | – | 0.996 |
| Gaussian jitter | 0.11 | 0.09 | 100% | 0.985 |
| SMOTE | 0.24 | 0.07 | 99% | 0.983 |
| WGAN-GP | 0.97 | 0.65 | 63% | 0.983 |
| Background (300 random points) | 1.53 | – | – | 0.500 |

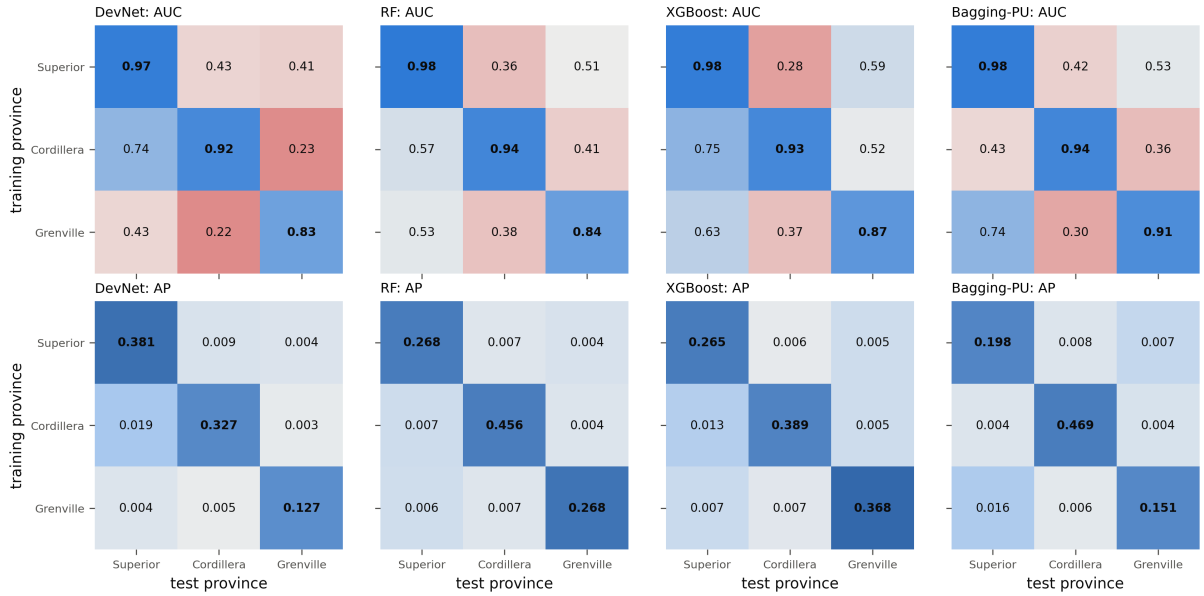



**Fig. 10** Transfer between provinces. Top: AUC; bottom: AP, coloured by AP divided by the test province's prevalence (log scale). Rows: province whose model (inputs, capacity, occurrences, training background) is applied; columns: province whose occurrences and evaluation background are scored (median of six seeds). The diagonal is random cross-validation within the province (Table 3).

### 4.2.5 Transferability between provinces

A province's model does not predict another province (Fig. 10). Within each province, random cross-validation gives an AP of 0.13–0.47 (nnPU excluded); applied to another province, the tree ensembles reach 0.004–0.016, about the random-ranking values of 0.004–0.009, and in 18 of the 30 model–province pairs the AUC is below 0.5: the model ranks the other province's occurrences below its background. The exceptions are DevNet, XGBoost and nnPU trained in Cordillera and bagging-PU trained in Grenville, applied to Superior: they reach AUCs of 0.73–0.75 but APs of only 0.013–0.086, against 0.005 for a random ranking. Leaving a province out of a national model gives the same picture: held-out Grenville is ranked close to chance (AUC 0.36–0.67, AP $\leq$ 0.037, against 0.004 for a random ranking), and held-out Superior reaches an AP of at most 0.051 (random ranking 0.005).

The provinces are distinct domains in the inputs. A random forest separates the training background of any two province groups with an AUC of 0.999–1.000, and the occurrences of Superior, Cordillera and Grenville from one another with 0.99–1.00. Several inputs mark occurrences in opposite directions in different provinces. Taken alone, LAB depth separates Superior occurrences from their background with an AUC of 0.22 (shallower LAB at occurrences) but Cordilleran ones with 0.67 (deeper), Moho depth with 0.35 and 0.78, and distance to regional greenstone with 0.23 and 0.63; deep-source magnetics marks Cordilleran and Grenville occurrences in opposite directions (0.25 and 0.81). This confirms that the provinces have to be modelled separately.

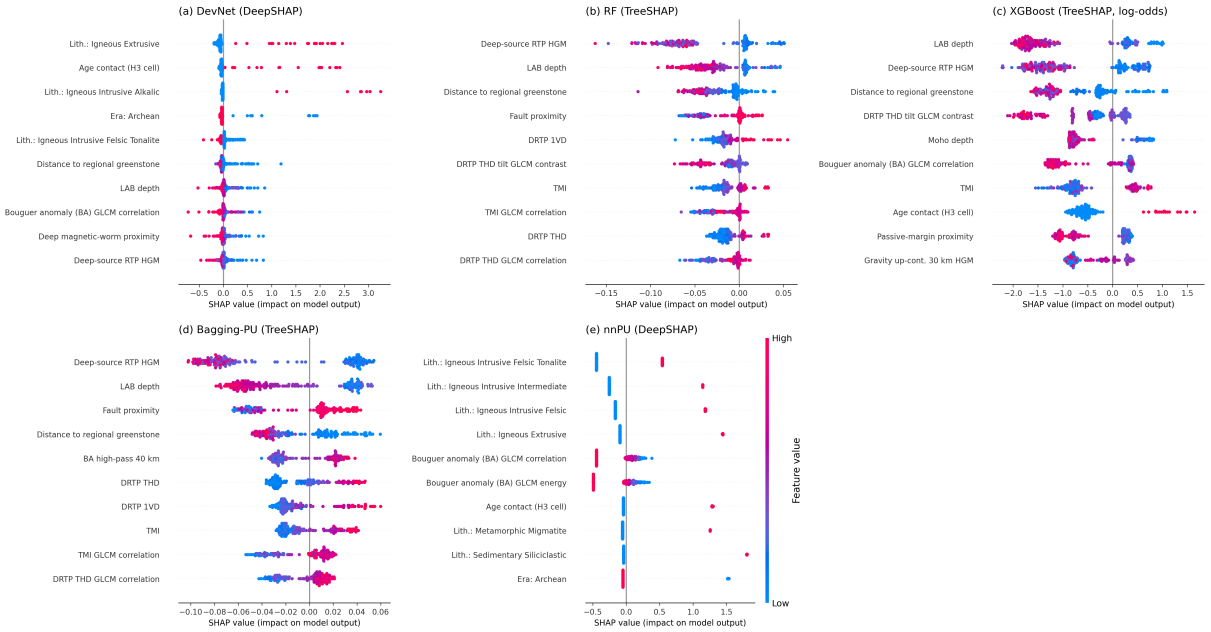



**Fig. 11** SHAP summaries of the production models for Superior, ten most important features each: (a) DevNet (DeepSHAP on its quantile-transformed inputs, mean over the five ensemble members), (b) RF (TreeSHAP), (c) XGBoost (TreeSHAP, log-odds), (d) bagging-PU (TreeSHAP, mean over the 200 trees) and (e) nnPU (DeepSHAP, as for DevNet). Each point is one explained sample; colour is the feature value. SHAP magnitudes are not comparable between models (different output scales).

### 4.2.6 Model interpretation

With the rank-based direction rule, RF and DevNet agree on the sign of the effect for all ten of their jointly most important features in each province (Fig. 11; Appendix C). Read from the sign of the mean SHAP value instead, the two models would appear to disagree on many of these features, an artefact of the two explainers' different baselines (Section 3.8).

The two models agree on directions but weight the evidence differently (Fig. 11a, b). In Superior, DevNet ranks five geological classes of the cell first: extrusive (greenstone) lithology, an age contact within the cell and alkalic intrusive lithology raise the score, and Archean era and tonalite lower it; it then ranks proximity to regional greenstone, LAB depth and Bouguer GLCM correlation (lower values more prospective). RF ranks the horizontal-gradient magnitude of deep-source magnetics first (lower values more prospective), then LAB depth, proximity to regional greenstone, fault proximity and the first vertical derivative of the reduced-to-pole magnetics, and gives the geological classes low ranks (22nd–41st of 57). Both models score ground as more prospective where the LAB is shallower, close to greenstone belts and faults, where TMI is high, and farther from deep magnetic worms. XGBoost and bagging-PU lead with the same inputs as RF, the horizontal-gradient magnitude of deep-source magnetics, LAB depth and distance to regional greenstone among their first four, with the same directions as RF and DevNet, whereas nnPU, the weakest model of the benchmark, relies mainly on lithology classes of the cell (Fig. 11c–e). In Cordillera both models treat Mesozoic terranes and ground far from high-grade metamorphic units as unfavourable and Proterozoic terranes as favourable. In Grenville both rank deep-source magnetics and the GLCM contrast of the Bouguer THD tilt among their five most important features, but given the weak predictive skill there these attributions describe the signal the models attempted to use rather than a reliable control.

### 4.2.7 Prospectivity maps and external validation

The production DevNet and RF maps (Fig. 12a, b) both concentrate their highest classes in the southern Superior Province, around the eastern-Abitibi cluster and across the Abitibi and Wawa subprovinces, but their rankings of the province are only moderately similar (Spearman correlation over the grid 0.33). RF assigns a probability of exactly zero to 80% of the grid, so its classes are drawn from the remaining

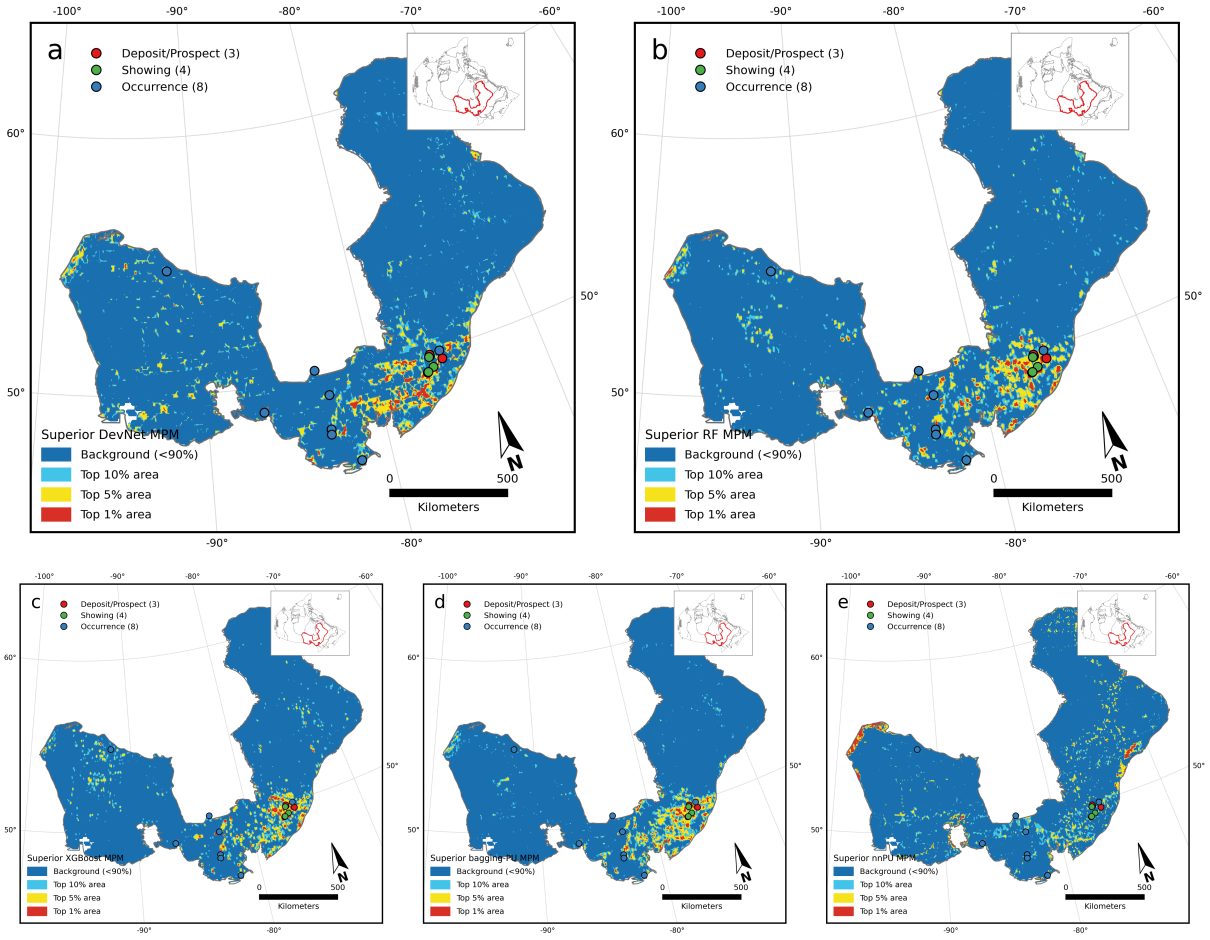



**Fig. 12** Superior prospectivity maps from the production (a) DevNet, (b) RF, (c) XGBoost, (d) bagging-PU and (e) nnPU models, classified by score percentile over the mapped area (top 1%, 5% and 10%), with known occurrences by development class (Canada Atlas Lambert). Scores on the 0.1° prediction grid were interpolated to a 2 km raster.

nodes. XGBoost and bagging-PU (Fig. 12c, d) also concentrate their highest classes in the southern Superior Province around the eastern-Abitibi cluster (Spearman correlation with DevNet over the grid 0.35 and 0.37), whereas nnPU (Fig. 12e), the weakest model of the benchmark, places most of its top 1% along the north-western and eastern margins of the province, away from the occurrences (0.16). In the Kapuskasing area (Fig. 13) DevNet places its top 1% mainly in the Abitibi Subprovince east of the KSZ; along the KSZ itself, the top 5% and 10% classes form discontinuous north-north-east trending patches on its western side. The largest top-1% patch in this frame, north of Timmins, lies in the greenstone of the western Abitibi Subprovince around the Kidd Creek Cu–Zn–Ag massive sulphide deposit (Hannington et al., 1999); no REE occurrence is recorded there, and the patch follows the greenstone setting that DevNet favours (Section 5.7). Clay-Howells, the untrained Fe–REE–Nb resource on the KSZ, lies 79 km from the nearest training occurrence; among the province's 8,000 random background points the production models place it at the 96.6th (DevNet), 99.8th (RF), 99.7th (XGBoost), 94.4th (bagging-PU) and 71.3rd (nnPU) percentile.

# 5 Discussion

## 5.1 National-scale models and transferability

The national experiments and the transfer tests explain why we model the provinces separately. First, the high test scores of the S2-grid design do not measure prediction on unexplored ground. Random splitting of augmented child cells leaves near-copies of the test positives in the training set, and negatives drawn from the least similar ground remove from evaluation exactly the ground on which a prospectivity model

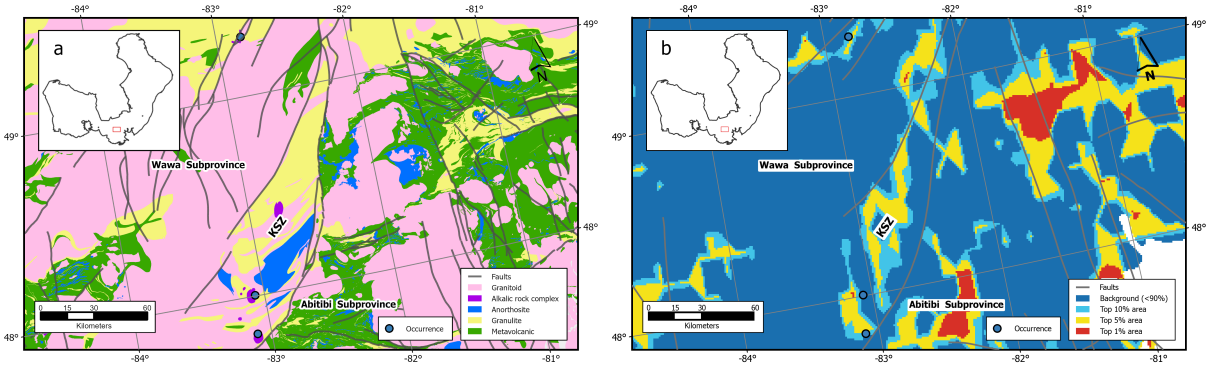



**Fig. 13** Kapuskasing area. (a) Regional geology with the Kapuskasing Structural Zone (KSZ) and alkalic rock complexes (Gittins et al., 1967; Percival & West, 1994), compiled from Heather et al. (1995) and Montsion et al. (2018). (b) DevNet top 1%, 5% and 10% classes (province-wide percentiles), with faults from the compilation used as an evidence layer, on the same frame as (a).

is hard to get right; either element alone gives a near-perfect AUC, and with one label per occurrence and random background at a realistic prevalence the same forest reaches an AP of 0.29. Second, a larger, pooled label set does not buy transferability. The provinces are separable domains, several layers mark occurrences in opposite directions (LAB and Moho depth, distance to greenstone), and models trained in one province rank another province's occurrences close to or below chance. In the terms of Meyer and Pebesma (2021), a province lies outside the area of applicability of a model trained on another, and pooling widens the training domain without making the model applicable to each part of it. A national model has to learn several expressions of the mineral system at once from 66 occurrences; a provincial model has to learn one.

## 5.2 Performance of the deviation network

In Superior DevNet ranks held-out occurrences better than RF, XGBoost and bagging-PU in every repeat of every protocol, and than nnPU in all but two blocked repeats, and its lead is largest when the test requires extrapolation: RF, XGBoost and bagging-PU, which learn sharp decision boundaries around the training positives, lose 33–68% of their AP once the nearest training occurrence is 50–200 km away, whereas DevNet keeps most of its AP to 100 km. With 15 positives the size of the margin over RF is uncertain (its 95% intervals reach down to 0.03), and DevNet's design was chosen with these provinces in view; but the lead is consistent across all 18 repeat–block combinations.

DevNet's advantage does not hinge on its two design choices (Appendix D, Table D6). With z-scored inputs, label-balanced batches or ten times more steps it still ranks held-out Superior occurrences well above RF under every protocol (AP 0.369–0.417 under random splits and 0.290–0.342 with 100 km blocks, against 0.268 and 0.129 for RF), and the final configuration is the best variant only with 100 and 200 km blocks (0.345 and 0.221). What the variants share is the deviation loss and a score that is a sum of a few ReLU ridges in the space of the inputs. A smooth score of this kind is rewarded when held-out occurrences lie far from the training ones, whereas the thresholds of tree ensembles carve out the neighbourhood of each training positive.

Among the learners that treat the background as unlabelled, the deviation loss is the one that works. nnPU, which shares DevNet's network and inputs and differs only in its risk, trails it by 0.13–0.30 in AP under blocking, fails in Cordillera and Grenville and without map classes, and depends on a class prior that is unknown in practice (AP 0.145–0.275 for $\pi$ between 0.005 and 0.05); bagging-PU trails it under every protocol.

The Cordilleran and Grenville results bound the conclusion. Where positives are comparatively plentiful and clustered (Cordillera), or the geophysical signal is weak (Grenville), the tree-based learners are as good or better under random splits, and no model transfers between blocks. DevNet is therefore not a universal replacement for discriminative baselines; its advantage is specific to the combination of very few positives, a shared geological setting and the need to extrapolate within it.

## 5.3 Geological map classes: evidence and lookup

The map classes move every model more than the choice of model does. Withholding them costs DevNet and RF about half or more of their AP, and a single class, alkalic intrusive rock, carries 30–49% of DevNet's blocked AP. Six of the 15 Superior occurrences lie on cells mapped as alkalic intrusive rock, which covers 0.5% of the background; for these occurrences the map already marks the target. The age-contact flag of the containing cell partly does the same, because a Proterozoic complex mapped within Archean crust creates an age contact, and DevNet ranks it second.

Whether such lookup is acceptable depends on the use. For targeting within mapped complexes, or for ranking known complexes by their geophysical setting, a mapped alkaline intrusion is legitimate and strong evidence, and the full inputs are appropriate. For finding complexes that the 1:5,000,000 map does not show, it is not, and the relevant skill is that without map classes: an AP of 0.075–0.123 under spatial blocking, 15–25 times the random-ranking value but well below what the full inputs suggest. Comparisons between studies, and claims about how well a model "discovers" occurrences, should state which of the two is being measured. In both settings DevNet is at or near the top under blocking; its absolute skill, however, depends strongly on the map classes.

## 5.4 Differences between provinces

Grenville is the weakest of the three provinces for every model and Superior the strongest. The evidence layers are the same national compilations everywhere and almost complete (less than 0.01% of values missing in each province), so the contrast does not reflect data coverage. It follows instead from the label sets and from the geology they sample.

First, the occurrences are less independent than their count suggests, most of all in Cordillera and Grenville: there, more than half of the occurrences lie within 10 km of another (median distance to the nearest other occurrence 7 km, against 31 km in Superior). Under a random split such neighbours fall on both sides, which is why the random-split skill in these two provinces is high and collapses under blocking (Section 4.2.2).

Second, the Grenville label set mixes carbonatites of different ages and tectonic settings. Crevier is a late-Grenvillian alkaline–carbonatite intrusion of about 957 Ma (Groulier et al., 2020). St-Honoré, with Terres Rompues on the same complex, was emplaced at about 580 Ma in the Saguenay graben of the St. Lawrence rift system (Néron et al., 2018). The Lavergne–Springer occurrences lie near Lake Nipissing at the western end of the Ottawa–Bonnechère graben, and the Cantley, Quinnville and Templeton showings near Gatineau belong to the calcite-rich veins and dykes of the Grenville marble belts, whose magmatic origin has long been debated (Moecher et al., 1997). Age diversity alone is not the problem, since the Superior occurrences also span several magmatic episodes (Rukhlov & Bell, 2010). They were, however, all emplaced into the same stable cratonic lithosphere, whose margins and crustal-scale structures the present-day LAB, Moho and potential-field layers still record. The Grenville crust was reworked by the Grenvillian orogeny and by later rifting, so its present-day geophysics need not record a common setting for occurrences that formed in different ones. The mapped geology adds noise of its own: the twelve Grenville occurrences fall on Archean, Proterozoic and Paleozoic map units, because the bedrock map records reworked protoliths and the Paleozoic cover over St-Honoré rather than the carbonatites, and none of them lies on a cell mapped as alkalic intrusive rock, whereas about 5% of the Grenville background does. The same lack of a common setting, at a larger scale, is why models do not transfer between provinces (Section 4.2.5).

Missing or coarse geophysics therefore explains little of the contrast; the number, independence and geological coherence of the labels explain most of it. A model for Grenville would need either more occurrences of one carbonatite type or separate models for rift-related and orogenic types, which the present label set cannot support.

## 5.5 Positive augmentation and label scarcity

Positive augmentation is an attractive response to label scarcity, and GAN-generated labels have been recommended for exactly this commodity (Parsa et al., 2025). Our results counsel caution. A generator fitted to a dozen training positives has two options. It can stay close to them, in which case it adds near-duplicates that carry no new information (jitter and SMOTE, nearest-neighbour ratios of 0.1–0.2); or it can spread out, in which case it must guess where unseen occurrences lie and, having no information about that, places part of its mass in background-like feature space (WGAN-GP, 37% of samples nearest

to a background point). Neither option supplies the information that is actually missing, namely what undiscovered occurrences look like. The gains for RF are better explained as smoothing: augmented points around and between the training positives make more trees vote for held-out points near them, which relieves RF's degenerate surface of exactly-zero probabilities rather than teaching it new geology. The WGAN-GP samples, which spread furthest, help RF most, also with 200 km blocks, yet even the best augmented RF stays well below DevNet without augmentation (0.191 against 0.345 with 100 km blocks), and DevNet, whose loss already produces smooth scores, gains little and inconsistently from the same points. Where positive augmentation is used, the augmented positives must be generated inside each training fold from that fold's positives only, and performance must be measured on real held-out occurrences; otherwise an augmented copy of a training occurrence in the test set reproduces the sibling problem of Section 4.1 in another form.

## 5.6 Validation design and reported performance

Four of our findings concern evaluation rather than models. First, with tens of positives a single cross-validation split gives a noisy estimate: re-splitting the same Superior data moves each model's AP over a range of 0.06–0.11 (Table 3), as large as many of the differences between models. Second, random splits overstate skill wherever occurrences cluster, and the size of the overstatement differs between models, so random splits can also reverse a model ranking (Section 4.2.2). Third, rebuilt nationally, the S2-grid design gives a test AUC of 1.000 with both of its classifiers, and two of its elements inflate it: drawing negatives from the least similar ground, which suffices on its own (0.998 and 0.996 without augmentation); and splitting augmented child cells at random, because almost every test positive has a sibling, sharing the coarse deep-architecture values and the map classes exactly and the fine-scale values nearly, on the training side (0.997 for the forest and 0.937 for the CNN with random negatives, against 0.923 and 0.880 with siblings kept together). In this design the held-out test asks the model to recognise a location it has already seen, and the comparison of training and test AUC, which Parsa et al. (2024) used to argue that their models were not over-fitted, becomes uninformative because the test set is not independent of the training set. Part of the gain that Parsa et al. (2024) report for augmentation is this leak: on our data, augmentation raises the balanced test AUC of the CNN from 0.824 to 0.937 under a random split but only to 0.880 when siblings are kept together (forest: 0.882, 0.997 and 0.923). Selecting negatives from the least similar 5% of the ground adds a second, independent source of optimism: it removes from evaluation exactly the ground on which a prospectivity model is hardest to get right. Fourth, the geological map classes can supply a large share of the reported skill by marking known complexes (Section 5.3), which no split design removes.

We stress what the national experiment does and does not show. It uses our inputs and labels with the two classifiers of Parsa et al. (2024), completed by settings that study does not report and without its grid search for the forest, and it does not establish what their numbers would have been under a grouped split. It does show that, on national Canadian carbonatite data, the published design produces a test AUC of 1.000 with either classifier, whereas with one label per occurrence, random background and a realistic prevalence the forest reaches an AUC of 0.903 and an AP of 0.292 and the CNN an AUC of 0.852 and an AP of 0.059, and that a model trained in one province does not carry over to another. Neither the high AUC nor its small spread over splits can be taken as evidence of generalisation to unexplored ground. Parsa et al. (2025) describe the same stratified random 70/30 split for a workflow that adds GAN-generated labels; their description does not state whether augmented labels derived from an original label were kept on the same side of the split as that label. If they were not, the caveat applies with greater force. For prospectivity studies under extreme label scarcity we therefore recommend (i) grouping every augmented or near-duplicate label with its source occurrence before any split, (ii) drawing evaluation negatives at random from the domain rather than from the ground least similar to the positives, (iii) reporting AP with intervals over repeated splits rather than a single AUC, (iv) adding a spatially blocked validation in which training background is purged from held-out blocks, (v) reporting results with and without point-level map classes, so that readers can see how much of a score is map lookup, and (vi) for models pooled across geological provinces, testing transfer to held-out provinces and computing extent-dependent inputs within fixed geological units.

### 5.7 Geological interpretation

The drivers on which RF and DevNet agree in Superior describe a coherent setting. Shallower LAB, together with the strong ranking of deep-source magnetic gradients, points to the thinner lithosphere at the edges of the thick Superior root, where small-degree melting of metasomatised mantle and access along lithosphere-scale structures are most likely (Griffin et al., 2013; Groves & Santosh, 2021; Hoggard et al., 2020); a mantle conductor beneath the southern Abitibi is consistent with such metasomatism (Snyder et al., 2021). Fault proximity and high magnetic intensity capture the structures that focus ascent and the magnetite-bearing intrusions themselves (Thomas et al., 2016). The negative effect of proximity to deep magnetic worms is less intuitive; it may reflect the concentration of deep magnetic contacts along terrane boundaries that are not themselves carbonatite hosts, and should not be over-interpreted.

The strong positive effect of greenstone lithology and of proximity to regional greenstone units needs more care. Eight of the 15 occurrences belong to the eastern-Abitibi cluster in a greenstone-dominated subprovince, so the association may partly record where the largest cluster lies rather than a control shared by the dispersed occurrences; the dispersed occurrences that DevNet ranks lowest under blocking, Kapuskasing and Nemegosenda Lake, lie in the high-grade and alkaline terrain of the KSZ rather than in a greenstone belt. Consistent with this, DevNet's map follows the KSZ only in discontinuous patches, while the untrained Clay-Howells resource on the KSZ still ranks at its 97th percentile. A model whose top classes are tied to one greenstone-hosted cluster is useful for ranking ground of that type, but it should not be expected to recognise alkaline complexes in other settings as well.

### 5.8 Limitations

Fifteen positives allow only weak statistical statements: our firm results are those that are consistent across repeats, block sizes, comparisons and provinces. DevNet's input representation and batch composition were chosen with diagnostic experiments on these three provinces, although with the present inputs every variant ranks Superior occurrences above RF (Appendix D); confirmation on an independent province is needed, and the Churchill Province, the obvious candidate, holds too few carbonatite-hosted occurrences after deduplication for a meaningful test. The geological encoding used here replaced a coarser grouping of lithologies after that grouping had been evaluated; the change raised the AP of every model except nnPU under random splits and 50–100 km blocks but not with 200 km blocks. The evaluation background may contain undiscovered occurrences, so all metrics are lower bounds. Radiometric data could not be used because of incomplete coverage, and whole-rock geochemistry is too sparse to serve as an input. The reduction to the pole assumes induced magnetisation; remanently magnetised bodies are not reduced correctly. The provenance of the fault compilation should be confirmed against the original data release before publication. The occurrence-class information (Deposit/Prospect, Showing, Occurrence) was not used for weighting; with 3–8 occurrences per class, confidence-weighted training could not be evaluated reliably. Finally, maps are per-region percentiles and are not calibrated probabilities. At the national scale, 12 of the 66 occurrences lie outside the three study provinces and four of them form one cluster (Ashram), so the "rest of Canada" group is small and spatially dependent; the national labels come from two compilations: a third (Huston, 2024), used by a parallel national study, would add one deposit (Hoidas Lake), and the five records of Orris et al. (2018) whose carbonatite affinity is marked as uncertain, which we excluded, would add five more (one of them Clay-Howells, our external check); and the 0.1° lattice samples high latitudes more densely per unit area.

## 6 Conclusions

For carbonatite-hosted REE prospectivity mapping in Canada, with 66 occurrences nationally and 12–27 per province:

1. Rebuilt on these data, the national S2-grid design of Parsa et al. (2024) gives a test AUC of 1.000 with both its random forest and its CNN; negatives drawn from the least similar ground suffice for it on their own (0.998 and 0.996 without augmentation), and random splitting of augmented child cells alone lifts the forest to 0.997. With one label per occurrence, random background and a realistic prevalence, the forest reaches an AUC of 0.903 and an AP of 0.292 and the CNN an AUC of 0.852 and an AP of 0.059.
2. In the Superior Province a semi-supervised Deviation Network ranks held-out occurrences better than Random Forest, XGBoost, bagging-PU and nnPU under random and 50–200 km spatially blocked

cross-validation (AP 0.38 and 0.37–0.22 against 0.27 and 0.18–0.09 for RF and XGBoost), in every repeat against RF, XGBoost and bagging-PU; with 15 occurrences the margin over RF is consistent but its size is uncertain. The advantage does not depend on DevNet's input scaling or batch composition.

3. Geological map classes move every model more than the choice of model. A single mapped alkalic-intrusion class, covering 6 of 15 Superior occurrences and 0.5% of the background, carries 30–49% of DevNet's blocked AP, and without point-level map classes DevNet and RF lose about half or more of their skill. Prospectivity studies should report results with and without such classes.
4. In the Cordilleran Orogen and the Grenville Province the tree-based learners lead under random splits and no model transfers between blocks; the number, independence and geological coherence of the labels, not data coverage, explain the contrast.
5. Positive augmentation with jitter, SMOTE or a tabular WGAN-GP, generated strictly within training folds, gave DevNet no consistent gain and raised RF's AP, most with WGAN-GP, without bringing it close to unaugmented DevNet.
6. Models do not transfer between geological provinces: a province's model ranks another province's occurrences at or below chance in most cases, the provinces are separable in the evidence layers, and several layers mark occurrences in opposite directions in different provinces. Prospectivity models for these occurrences should therefore be built per province.
7. RF and DevNet agree on the direction of all 30 leading drivers across the three provinces. DevNet's Superior map places the untrained Clay-Howells Fe–REE–Nb resource at its 97th percentile and favours greenstone-hosted settings like the dominant Abitibi cluster.

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## Declarations

**Funding.** Not applicable.

**Competing interests.** The authors declare no competing interests.

**Data availability.** Occurrence labels are derived from Orris et al. (2018) and Parsa et al. (2024); the deduplicated provincial and national tables and the study-area outlines are distributed with the code. All evidence layers are public and available from McCafferty et al. (2023), Jobin et al. (2017), Miles and Oneschuk (2016), D.C. Bradley (2008) and Wheeler et al. (1996); the derived gravity and magnetic grids are reproduced by the code.

**Code availability.** The pipeline, every analysis in this paper and the scripts that generate its tables and figures are available at https://github.com/miaozhou-312/ree-prospectivity-paper.

## Appendix A Additional results

**Table A1** AUC (median over six repeats) for the runs of Table 4. Bold: highest AUC in each row.

| Region | Split | DevNet | RF | XGBoost | Bagging-PU | nnPU |
|---|---|---|---|---|---|---|
| Superior | random | 0.972 | **0.985** | 0.976 | 0.976 | 0.842 |
| | 50 km | 0.969 | **0.976** | 0.962 | 0.964 | 0.862 |
| | 100 km | **0.928** | 0.922 | 0.904 | 0.921 | 0.846 |
| | 200 km | **0.886** | 0.861 | 0.826 | 0.837 | 0.819 |
| Cordillera | random | 0.925 | 0.936 | 0.926 | **0.937** | 0.575 |
| | 50 km | 0.866 | 0.856 | 0.840 | **0.879** | 0.581 |
| | 100 km | 0.747 | 0.841 | 0.782 | **0.856** | 0.565 |
| | 200 km | 0.744 | 0.720 | 0.712 | **0.825** | 0.584 |
| Grenville | random | 0.826 | 0.836 | 0.870 | **0.909** | 0.609 |
| | 50 km | 0.755 | 0.717 | 0.777 | **0.825** | 0.620 |
| | 100 km | 0.699 | 0.687 | 0.722 | **0.780** | 0.598 |
| | 200 km | 0.681 | 0.582 | 0.665 | **0.691** | 0.609 |

**Table A2** Sensitivity of nnPU to the assumed class prior $\pi$, Superior (random cross-validation; median and 95% interval over six repeats).

| $\pi$ | AUC | AP |
|---|---|---|
| 0.005 | 0.823 [0.761, 0.860] | 0.145 [0.079, 0.177] |
| 0.01 | 0.842 [0.772, 0.873] | 0.159 [0.096, 0.210] |
| 0.02 | 0.864 [0.827, 0.897] | 0.195 [0.115, 0.251] |
| 0.05 | 0.885 [0.878, 0.919] | 0.275 [0.186, 0.395] |

# Appendix B Positive augmentation: generators and spatial blocking

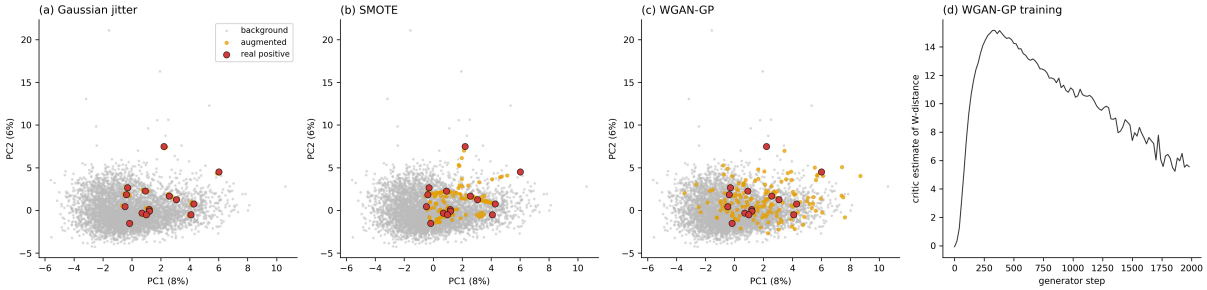



**Fig. B1** (a–c) Augmented positives from each generator (fitted to all 15 Superior positives) in the first two principal components of background and real positives. (d) WGAN-GP critic estimate of the Wasserstein distance during training.

**Table B3** Superior average precision with fold-safe positive augmentation under spatially blocked cross-validation (median and 95% interval over six repeats). DevNet's runs were computed on a second machine; its training is not bit-for-bit identical across machines, so its unaugmented values differ from Table 4 in the third decimal.

| Blocks | Model | Augmentation | AP |
|---|---|---|---|
| 100 km | DevNet | none | 0.343 [0.163, 0.414] |
| | | Gaussian jitter 10× | 0.341 [0.156, 0.420] |
| | | SMOTE 3× | 0.324 [0.146, 0.418] |
| | | WGAN-GP 3× | 0.326 [0.228, 0.398] |
| | | WGAN-GP 10× | 0.308 [0.172, 0.350] |
| 100 km | RF | none | 0.129 [0.067, 0.161] |
| | | Gaussian jitter 10× | 0.139 [0.074, 0.187] |
| | | SMOTE 3× | 0.110 [0.053, 0.168] |
| | | WGAN-GP 3× | 0.135 [0.074, 0.202] |
| | | WGAN-GP 10× | 0.191 [0.083, 0.294] |
| 200 km | DevNet | none | 0.216 [0.154, 0.310] |
| | | Gaussian jitter 10× | 0.238 [0.157, 0.314] |
| | | SMOTE 3× | 0.200 [0.155, 0.251] |
| | | WGAN-GP 3× | 0.246 [0.165, 0.314] |
| | | WGAN-GP 10× | 0.258 [0.171, 0.305] |
| 200 km | RF | none | 0.087 [0.053, 0.149] |
| | | Gaussian jitter 10× | 0.096 [0.062, 0.137] |
| | | SMOTE 3× | 0.077 [0.047, 0.118] |
| | | WGAN-GP 3× | 0.132 [0.080, 0.199] |
| | | WGAN-GP 10× | 0.122 [0.074, 0.256] |

# Appendix C Interpretation in the contrasting provinces

**Table C4** Cordillera: the ten jointly most important features of DevNet and RF (mean |SHAP|, normalised per model). Rank among 61 features; direction ↑/↓: higher values (or presence) raise/lower the score, with Spearman $\rho$ between feature value and SHAP value.

| Feature | DevNet rank | DevNet dir. | RF rank | RF dir. |
|---|---|---|---|---|
| Distance to regional high-grade metam. | 5 | ↓ (-0.96) | 1 | ↓ (-0.86) |
| Era: Proterozoic | 2 | ↑ (+0.59) | 19 | ↑ (+0.70) |
| Era: Mesozoic | 4 | ↓ (-0.82) | 11 | ↓ (-0.82) |
| Bouguer anomaly (BA) | 9 | ↓ (-0.96) | 3 | ↓ (-0.87) |
| Age contact (H3 cell) | 1 | ↑ (+0.84) | 44 | ↑ (+0.84) |
| DRTP THD GLCM correlation | 3 | ↓ (-0.82) | 27 | ↓ (-0.42) |
| Moho depth | 8 | ↑ (+0.96) | 6 | ↑ (+0.91) |
| Distance to regional greenstone | 17 | ↑ (+0.93) | 4 | ↑ (+0.34) |
| Passive-margin proximity | 41 | ↑ (+0.28) | 2 | ↑ (+0.81) |
| Distance to regional age contact | 7 | ↑ (+0.95) | 14 | ↑ (+0.92) |

**Table C5** Grenville: the ten jointly most important features of DevNet and RF (mean |SHAP|, normalised per model). Rank among 66 features; direction ↑/↓: higher values (or presence) raise/lower the score, with Spearman $\rho$ between feature value and SHAP value.

| Feature | DevNet rank | DevNet dir. | RF rank | RF dir. |
|---|---|---|---|---|
| Distance to regional age contact | 9 | ↓ (-0.96) | 1 | ↓ (-0.79) |
| Deep-source RTP | 5 | ↑ (+0.93) | 2 | ↑ (+0.73) |
| DRTP THD tilt | 10 | ↓ (-0.95) | 3 | ↓ (-0.83) |
| DRTP THD tilt GLCM contrast | 1 | ↓ (-0.94) | 10 | ↓ (-0.93) |
| BA THD tilt GLCM contrast | 4 | ↑ (+0.97) | 5 | ↑ (+0.83) |
| Distance to regional mafic-ultramafic | 24 | ↑ (+0.92) | 4 | ↑ (+0.42) |
| Distance to regional sedimentary | 15 | ↓ (-0.95) | 6 | ↓ (-0.90) |
| DRTP mean (10 km) | 6 | ↑ (+0.95) | 9 | ↑ (+0.93) |
| Moho depth | 22 | ↓ (-0.92) | 7 | ↓ (-0.69) |
| Deep magnetic-worm proximity | 3 | ↑ (+0.97) | 25 | ↑ (+0.79) |

# Appendix D Ablation of DevNet design choices

DevNet's two design choices (Section 3.3) are examined here with the inputs of the main analysis, by changing one element at a time from a configuration with z-scored inputs and label-balanced batches of 16 + 16 rows. All configurations are evaluated with the folds, protocols and seeds of the main results (Table D6). In Superior the variants differ little, and none is best under every protocol: under random splits z-scored inputs with background-heavy batches give the highest AP (0.417 against 0.381 for the final configuration), under 50 km blocks label-balanced batches do (0.390 and 0.384 against 0.371), and with 100 and 200 km blocks the final configuration is highest (0.345 and 0.221). Every variant stays well above RF (0.268, 0.165, 0.129 and 0.087; Table 4). Training for longer repeats the dozen positives and pushes the network towards memorising their neighbourhoods: with z-scored inputs, ten times more steps raised random-split AP (0.401 against 0.369) but lowered it under 100 km blocks (0.290 against 0.329). The nested early-stopping configuration chooses the number of steps (100–3,000, cosine schedule over 3,000 steps) inside each outer training fold by a 100 km spatial cross-validation (three inner folds, two inner restarts, background purged from inner validation blocks), so the test fold is never used for the choice. Table D7 gives the sensitivity of the final configuration to the deviation margin: smaller margins gave higher AP in Superior and Cordillera (Superior 0.472 with $a = 2$ against 0.381 with $a = 5$) but not in Grenville, and we keep the value of Pang et al. (2019) rather than tune it on these folds.

**Table D6** Ablation of DevNet's design choices: AP (median over six repeats) under random cross-validation and spatial blocks, on identical folds. Inputs: quantile-normal or z-scored; batch: positive + background rows per step; steps: number of gradient steps (nested ES: chosen by nested 100 km spatial cross-validation). Bold: the final configuration. –: not run.

| Region | Inputs | Batch | Steps | random | 50 km | 100 km | 200 km |
|---|---|---|---|---|---|---|---|
| Superior | quantile | 16 + 240 | 300 (final) | **0.381** | **0.371** | **0.345** | **0.221** |
| | z-score | 16 + 240 | 300 | 0.417 | 0.378 | 0.342 | 0.212 |
| | quantile | 16 + 16 | 300 | 0.397 | 0.384 | 0.329 | 0.220 |
| | z-score | 16 + 16 | 300 | 0.369 | 0.390 | 0.329 | 0.201 |
| | z-score | 16 + 16 | 3,000 | 0.401 | – | 0.290 | – |
| | z-score | 16 + 16 | nested ES | 0.401 | 0.358 | 0.297 | 0.190 |
| Cordillera | quantile | 16 + 240 | 300 (final) | **0.327** | **0.131** | **0.048** | **0.056** |
| | z-score | 16 + 240 | 300 | 0.303 | 0.136 | 0.074 | 0.049 |
| | quantile | 16 + 16 | 300 | 0.307 | 0.124 | 0.045 | 0.052 |
| | z-score | 16 + 16 | 300 | 0.273 | 0.135 | 0.077 | 0.045 |
| | z-score | 16 + 16 | 3,000 | 0.255 | – | 0.060 | – |
| | z-score | 16 + 16 | nested ES | 0.284 | 0.140 | 0.069 | 0.047 |
| Grenville | quantile | 16 + 240 | 300 (final) | **0.127** | **0.036** | **0.008** | **0.007** |
| | z-score | 16 + 240 | 300 | 0.113 | 0.030 | 0.007 | 0.006 |
| | quantile | 16 + 16 | 300 | 0.113 | 0.031 | 0.008 | 0.008 |
| | z-score | 16 + 16 | 300 | 0.078 | 0.026 | 0.009 | 0.008 |
| | z-score | 16 + 16 | 3,000 | 0.129 | – | 0.010 | – |
| | z-score | 16 + 16 | nested ES | 0.107 | 0.022 | 0.008 | 0.008 |

**Table D7** Sensitivity of DevNet to the deviation margin $a$ (AP, median over six repeats, random cross-validation). The main results use $a = 5$.

| Region | $a = 2$ | $a = 3$ | $a = 5$ | $a = 8$ |
|---|---|---|---|---|
| Superior | 0.472 | 0.446 | 0.381 | 0.377 |
| Cordillera | 0.367 | 0.379 | 0.327 | 0.313 |
| Grenville | 0.126 | 0.115 | 0.127 | 0.120 |

## References


Agterberg, F.P., & Bonham-Carter, G.F. (2005). Measuring the performance of mineral-potential maps. *Natural Resources Research*, *14*(1), 1–17, https://doi.org/10.1007/s11053-005-4674-0

Airola, A., Pohjankukka, J., Torppa, J., Middleton, M., Nykänen, V., Heikkonen, J., Pahikkala, T. (2019). The spatial leave-pair-out cross-validation method for reliable AUC estimation of spatial classifiers. *Data Mining and Knowledge Discovery*, *33*(3), 730–747, https://doi.org/10.1007/s10618-018-00607-x

Alken, P., Thébault, E., Beggan, C.D., et al. (2021). International Geomagnetic Reference Field: the thirteenth generation. *Earth, Planets and Space*, *73*, 49, https://doi.org/10.1186/s40623-020-01288-x

Aranha, M., Porwal, A., Sundaralingam, M., González-Álvarez, I. (2021). Detection and extraction of geophysical features associated with carbonatites and alkaline complexes. *First international meeting for applied geoscience & energy expanded abstracts* (pp. 894–899).

Arjovsky, M., Chintala, S., Bottou, L. (2017). Wasserstein generative adversarial networks. *Proceedings of the 34th international conference on machine learning* (Vol. 70, pp. 214–223).

Arkani-Hamed, J. (1988). Differential reduction-to-the-pole of regional magnetic anomalies. *Geophysics*, *53*(12), 1592–1600, https://doi.org/10.1190/1.1442441

Bekker, J., & Davis, J. (2020). Learning from positive and unlabeled data: A survey. *Machine Learning*, *109*(4), 719–760, https://doi.org/10.1007/s10994-020-05877-5

Blakely, R.J. (1995). *Potential theory in gravity and magnetic applications.* Cambridge: Cambridge University Press.

Bradley, A.P. (1997). The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognition*, *30*(7), 1145–1159, https://doi.org/10.1016/S0031-3203(96)00142-2

Bradley, D.C. (2008). Passive margins through earth history. *Earth-Science Reviews*, *91*(1–4), 1–26, https://doi.org/10.1016/j.earscirev.2008.08.001

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32, https://doi.org/10.1023/A:1010933404324

Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, *16*, 321–357, https://doi.org/10.1613/jair.953

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining* (pp. 785–794).

Chen, Y., & Wu, W. (2019). Isolation forest as an alternative data-driven mineral prospectivity mapping method with a higher data-processing efficiency. *Natural Resources Research*, *28*(1), 31–46, https://doi.org/10.1007/s11053-018-9375-6

Chung, C.-J.F., & Fabbri, A.G. (2003). Validation of spatial prediction models for landslide hazard mapping. *Natural Hazards*, *30*(3), 451–472, https://doi.org/10.1023/B:NHAZ.0000007172.62651.2b

Coolbaugh, M.F., Raines, G.L., Zehner, R.E. (2007). Assessment of exploration bias in data-driven predictive models and the estimation of undiscovered resources. *Natural Resources Research*, *16*(2), 199–207, https://doi.org/10.1007/s11053-007-9037-6

Cooper, G.R.J., & Cowan, D.R. (2005a). Differential reduction to the pole. *Computers & Geosciences*, *31*(8), 989–999,

Cooper, G.R.J., & Cowan, D.R. (2005b). The use of textural analysis to locate features in geophysical data. *Computers & Geosciences*, *31*(7), 882–890, https://doi.org/10.1016/j.cageo.2005.02.001

Dormann, C.F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., … Lautenbach, S. (2013). Collinearity: A review of methods to deal with it and a simulation study evaluating their performance. *Ecography*, *36*(1), 27–46, https://doi.org/10.1111/j.1600-0587.2012.07348.x

Drenth, B.J. (2014). Geophysical expression of a buried niobium and rare earth element deposit: The Elk Creek carbonatite, Nebraska, USA. *Interpretation*, *2*(4), SJ23–SJ33, https://doi.org/10.1190/INT-2014-0002.1

Efron, B., & Tibshirani, R.J. (1993). *An introduction to the bootstrap*. New York: Chapman & Hall.

Elkan, C., & Noto, K. (2008). Learning classifiers from only positive and unlabeled data. *Proceedings of the 14th acm sigkdd international conference on knowledge discovery and data mining* (pp. 213–220).

Gittins, J., Macintyre, R.M., York, D. (1967). The ages of carbonatite complexes in eastern Canada. *Canadian Journal of Earth Sciences*, *4*(4), 651–655, https://doi.org/10.1139/e67-042

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., … Bengio, Y. (2014). Generative adversarial nets. *Advances in neural information processing systems 27* (pp. 2672–2680).

Gorishniy, Y., Rubachev, I., Khrulkov, V., Babenko, A. (2021). Revisiting deep learning models for tabular data. *Advances in neural information processing systems 34* (pp. 18932–18943).

Griffin, W.L., Begg, G.C., O'Reilly, S.Y. (2013). Continental-root control on the genesis of magmatic ore deposits. *Nature Geoscience*, *6*(11), 905–910, https://doi.org/10.1038/ngeo1954

Grinsztajn, L., Oyallon, E., Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in neural information processing systems 35, datasets and benchmarks track* (pp. 507–520).

Groulier, P.-A., et al. (2020). Silicate–carbonate liquid immiscibility: Insights from the Crevier alkaline intrusion (Quebec). *Journal of Petrology*, *61*(3), egaa033, https://doi.org/10.1093/petrology/egaa033

Groves, D.I., & Santosh, M. (2021). Craton and thick lithosphere margins: The sites of giant mineral deposits and mineral provinces. *Gondwana Research*, *100*, 195–222, https://doi.org/10.1016/j.gr.2020.06.008

Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., Courville, A. (2017). Improved training of Wasserstein GANs. *Advances in neural information processing systems 30* (pp. 5767–5777).

Hall-Beyer, M. (2017). Practical guidelines for choosing GLCM textures to use in landscape classification tasks over a range of moderate spatial scales. *International Journal of Remote Sensing*, *38*(5), 1312–1338, https://doi.org/10.1080/01431161.2016.1278314

Hannington, M.D., Barrie, C.T., Bleeker, W. (1999). The giant Kidd Creek volcanogenic massive sulfide deposit, western Abitibi Subprovince, Canada: Summary and synthesis. M.D. Hannington & C.T. Barrie (Eds.), *The giant Kidd Creek volcanogenic massive sulfide deposit, western Abitibi Subprovince, Canada* (Vol. 10, pp. 661–672). Society of Economic Geologists.

Heather, K.B., Percival, J.A., Moser, D., Bleeker, W. (1995). *Tectonics and metallogeny of Archean crust in the Abitibi–Kapuskasing–Wawa region* (Open File No. 3141). Geological Survey of Canada.

Hoggard, M.J., Czarnota, K., Richards, F.D., Huston, D.L., Jaques, A.L., Ghelichkhan, S. (2020). Global distribution of sediment-hosted metals controlled by craton edge stability. *Nature Geoscience*, *13*(7), 504–510, https://doi.org/10.1038/s41561-020-0593-2

Hornby, P., Boschetti, F., Horowitz, F.G. (1999). Analysis of potential field data in the wavelet domain. *Geophysical Journal International*, *137*(1), 175–196, https://doi.org/10.1046/j.1365-246x.1999.00788.x

Hou, Z.-Q., Xu, B., Zhang, H., Zheng, Y.-C., Wang, R., Liu, Y., … O'Reilly, S.Y. (2023). Refertilized continental root controls the formation of the Mianning–Dechang carbonatite-associated rare-earth-element ore system. *Communications Earth & Environment*, *4*(1), 293, https://doi.org/10.1038/s43247-023-00956-6

Huston, D. (2024). *Global rare earth element deposits – a compilation.* Geoscience Australia, Canberra.

Jobin, D.M., Véronneau, M., Miles, W. (2017). *Gravity anomaly map, Canada* (Canadian Geoscience Map No. 8081). Geological Survey of Canada.

Kingma, D.P., & Ba, J. (2014). *Adam: A method for stochastic optimization.* arXiv:1412.6980.

Kiryo, R., Niu, G., du Plessis, M.C., Sugiyama, M. (2017). Positive-unlabeled learning with non-negative risk estimator. *Advances in neural information processing systems 30* (pp. 1675–1685).

Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair, B., … Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, *2*(1), 56–67, https://doi.org/10.1038/s42256-019-0138-9

Lundberg, S.M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems 30* (pp. 4765–4774).

Luo, Z., Farahbakhsh, E., Hore, S., Müller, R.D. (2026). DEEP-SEAM: An explainable semi-supervised deep learning framework for mineral prospectivity mapping. *Geoscientific Model Development*, *19*(7), 2593–2625, https://doi.org/10.5194/gmd-19-2593-2026

McCafferty, A.E., San Juan, C.A., Lawley, C., Graham, G.E., Gadd, M.G., Huston, D.L., … Czarnota, K. (2023). *National-scale geophysical, geologic, and mineral resource data and grids for the United States, Canada, and Australia: Data in support of the Tri-National Critical Minerals Mapping Initiative (ver. 1.1, march 2025).* U.S. Geological Survey data release.

McCuaig, T.C., Beresford, S., Hronsky, J. (2010). Translating the mineral systems approach into an effective exploration targeting system. *Ore Geology Reviews*, *38*(3), 128–138, https://doi.org/10.1016/j.oregeorev.2010.05.008

Meyer, H., & Pebesma, E. (2021). Predicting into unknown space? estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, *12*(9), 1620–1633, https://doi.org/10.1111/2041-210X.13650

Miles, W., & Oneschuk, D. (2016). *Magnetic anomaly map, Canada* (Canadian Geoscience Map No. 7799). Geological Survey of Canada.

Miller, H.G., & Singh, V. (1994). Potential field tilt—a new concept for location of potential field sources. *Journal of Applied Geophysics*, *32*(2–3), 213–217, https://doi.org/10.1016/0926-9851(94)90022-1

Moecher, D.P., Anderson, E.D., Cook, C.A., Mezger, K. (1997). The petrogenesis of metamorphosed carbonatites in the Grenville Province, Ontario. *Canadian Journal of Earth Sciences*, *34*(9), 1185–1201, https://doi.org/10.1139/e17-095

Montsion, R.M., Thurston, P., Ayer, J. (2018). *1:2 000 000 scale geological compilation of the Superior Craton – version 1* (Tech. Rep. No. MERC-ME-2018-017). Sudbury, Ontario: Mineral Exploration Research Centre, Harquail School of Earth Sciences, Laurentian University.

Mordelet, F., & Vert, J.-P. (2014). A bagging SVM to learn from positive and unlabeled examples. *Pattern Recognition Letters*, *37*, 201–209, https://doi.org/10.1016/j.patrec.2013.06.010

Néron, A., Bédard, L.P., Gaboury, D. (2018). The Saint-Honoré carbonatite REE zone, Québec, Canada: Combined magmatic and hydrothermal processes. *Minerals*, *8*(9), 397, https://doi.org/10.3390/min8090397

Ontario Geological Survey (2011). *Mineral deposit inventory record MDI42G16SE00006: Clay-Howells.* Ontario Ministry of Northern Development and Mines. Retrieved from https://www.geologyontario.mines.gov.on.ca/mineral-inventory/MDI42G16SE00006

Orris, G.J., Seo, Y., Briggs, D.A., Dunlap, P., Cocker, M.D. (2018). *Global rare earth element occurrence database.* U.S. Geological Survey data release.

Pang, G., Shen, C., van den Hengel, A. (2019). Deep anomaly detection with deviation networks. *Proceedings of the 25th acm sigkdd international conference on knowledge discovery & data mining* (pp. 353–362).

Parker, B.J., Günter, S., Bedo, J. (2007). Stratification bias in low signal microarray studies. *BMC Bioinformatics*, *8*(1), 326, https://doi.org/10.1186/1471-2105-8-326

Parsa, M., & Carranza, E.J.M. (2021). Modulating the impacts of stochastic uncertainties linked to deposit locations in data-driven predictive mapping of mineral prospectivity. *Natural Resources Research*, *30*(5), 3081–3097, https://doi.org/10.1007/s11053-021-09891-9

Parsa, M., Lawley, C.J.M., Cawood, T., Martins, T., Cumani, R., Zhang, S.E., … Voinot, A. (2025). Pan-Canadian predictive modeling of lithium–cesium–tantalum pegmatites with deep learning and natural language processing. *Natural Resources Research*, *34*, 639–668, https://doi.org/10.1007/s11053-024-10438-x

Parsa, M., Lawley, C.J.M., Cumani, R., Schetselaar, E., Harris, J., Lentz, D.R., … Bourdeau, J.E. (2024). Predictive modeling of Canadian carbonatite-hosted REE +/– Nb deposits. *Natural Resources Research*, *33*(5), 1941–1965, https://doi.org/10.1007/s11053-024-10369-7

Percival, J.A., & West, G.F. (1994). The Kapuskasing uplift: A geological and geophysical synthesis. *Canadian Journal of Earth Sciences*, *31*(7), 1256–1286, https://doi.org/10.1139/e94-110

Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., … Pélissier, R. (2020). Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, *11*, 4540, https://doi.org/10.1038/s41467-020-18321-y

Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., … Dormann, C.F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, *40*(8), 913–929, https://doi.org/10.1111/ecog.02881

Rukhlov, A.S., & Bell, K. (2010). Geochronology of carbonatites from the Canadian and Baltic Shields, and the Canadian Cordillera: Clues to mantle evolution. *Mineralogy and Petrology*, *98*(1), 11–54, https://doi.org/10.1007/s00710-009-0054-5

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, *10*(3), e0118432, https://doi.org/10.1371/journal.pone.0118432

Simandl, G.J., & Paradis, S. (2018). Carbonatites: Related ore deposits, resources, footprint, and exploration methods. *Applied Earth Science*, *127*(4), 123–152, https://doi.org/10.1080/25726838.2018.1516935

Snyder, D.B., Savard, G., Kjarsgaard, B.A., Vaillancourt, A., Thurston, P.C., Ayer, J.A., Roots, E. (2021). Multidisciplinary modeling of mantle lithosphere structure within the Superior Craton, North America. *Geochemistry, Geophysics, Geosystems*, *22*(4), e2020GC009566, https://doi.org/10.1029/2020GC009566

Thomas, M.D., Ford, K.L., Keating, P. (2016). Review paper: Exploration geophysics for intrusion-hosted rare metals. *Geophysical Prospecting*, *64*(5), 1275–1304, https://doi.org/10.1111/1365-2478.12352

Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillera-Arroita, G. (2019). blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, *10*(2), 225–232, https://doi.org/10.1111/2041-210X.13107

Verduzco, B., Fairhead, J.D., Green, C.M., MacKenzie, C. (2004). New insights into magnetic derivatives for structural mapping. *The Leading Edge*, *23*(2), 116–119, https://doi.org/10.1190/1.1651454

Verplanck, P.L., Van Gosen, B.S., Seal II, R.R., McCafferty, A.E. (2014). *A deposit model for carbonatite and peralkaline intrusion-related rare earth element deposits* (Scientific Investigations Report No. 2010-5070-J). U.S. Geological Survey.

Wang, Z.-Y., Fan, H.-R., Zhou, L., Yang, K.-F., She, H.-D. (2020). Carbonatite-related REE deposits: An overview. *Minerals*, *10*(11), 965, https://doi.org/10.3390/min10110965

Wheeler, J.O., Hoffman, P.F., Card, K.D., Davidson, A., Sanford, B.V., Okulitch, A.V., Roest, W.R. (1996). *Geological map of Canada* (Map No. 1860A). Geological Survey of Canada.

Woolley, A.R., & Bailey, D.K. (2012). The crucial role of lithospheric structure in the generation and release of carbonatites: Geological evidence. *Mineralogical Magazine*, *76*(2), 259–270, https://doi.org/10.1180/minmag.2012.076.2.02

Xiong, Y., & Zuo, R. (2016). Recognition of geochemical anomalies using a deep autoencoder network. *Computers & Geosciences*, *86*, 75–82, https://doi.org/10.1016/j.cageo.2015.10.006

Xiong, Y., & Zuo, R. (2018). GIS-based rare events logistic regression for mineral prospectivity mapping. *Computers & Geosciences*, *111*, 18–25, https://doi.org/10.1016/j.cageo.2017.10.005

Xiong, Y., & Zuo, R. (2021). A positive and unlabeled learning algorithm for mineral prospectivity mapping. *Computers & Geosciences*, *147*, 104667, https://doi.org/10.1016/j.cageo.2020.104667

Yaxley, G.M., Kjarsgaard, B.A., Jaques, A.L. (2021). Evolution of carbonatite magmas in the upper mantle and crust. *Elements*, *17*(5), 315–320, https://doi.org/10.2138/gselements.17.5.315

Zuo, R., Kreuzer, O.P., Wang, J., Xiong, Y., Zhang, Z., Wang, Z. (2021). Uncertainties in GIS-based mineral prospectivity mapping: Key types, potential impacts and possible solutions. *Natural Resources Research*, *30*(5), 3059–3079, https://doi.org/10.1007/s11053-021-09871-z

Zuo, R., & Wang, Z. (2020). Effects of random negative training samples on mineral prospectivity mapping. *Natural Resources Research*, *29*(6), 3443–3455, https://doi.org/10.1007/s11053-020-09668-6